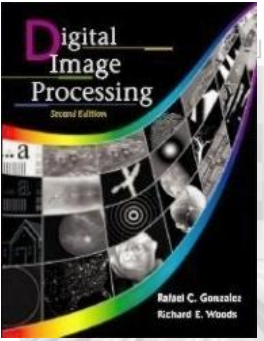


Chapter 4
Image Enhancement in the Frequency Domain

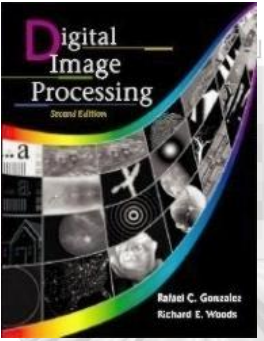
© 2002 R. C. Gonzalez & R. E. Woods



Background

- Any function that periodically repeats itself can be expressed as the sum of sines and/or cosines of different frequencies, each multiplied by a different coefficient (Fourier series).
- Even functions that are not periodic (but whose area under the curve is finite) can be expressed as the integral of sines and/or cosines multiplied by a weighting function (Fourier transform).

© 2002 R. C. Gonzalez & R. E. Woods



Background

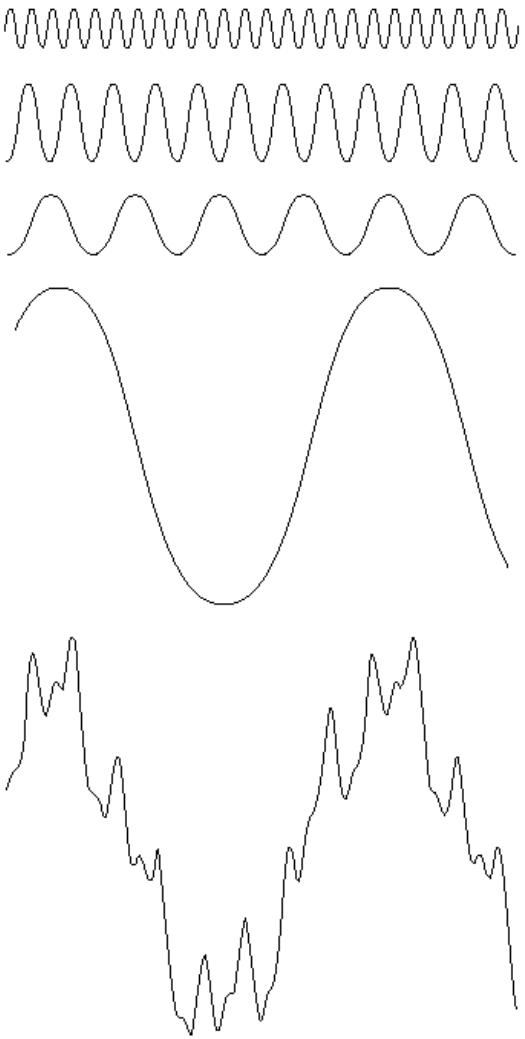
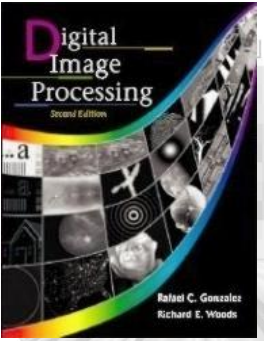


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

- The **frequency domain** refers to the plane of the two dimensional discrete Fourier transform of an image.
- The purpose of the Fourier transform is to represent a signal as a linear combination of sinusoidal signals of various frequencies.

© 2002 R. C. Gonzalez & R. E. Woods



Introduction to the Fourier Transform and the Frequency Domain

- The one-dimensional Fourier transform and its inverse
  - Fourier transform (continuous case)

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx \quad \text{where } j = \sqrt{-1}$$

  - Inverse Fourier transform:

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du$$

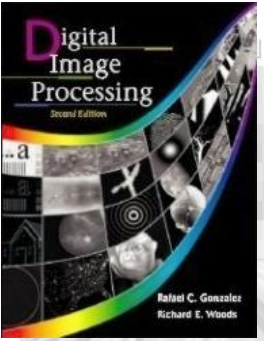
- The two-dimensional Fourier transform and its inverse
  - Fourier transform (continuous case)

$$F(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y) e^{-j2\pi(ux+vy)} dxdy$$

  - Inverse Fourier transform:

$$f(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(u,v) e^{j2\pi(ux+vy)} dudv$$

© 2002 R. C. Gonzalez & R. E. Woods



Introduction to the Fourier Transform and the Frequency Domain

- The one-dimensional Fourier transform and its inverse
  - Fourier transform (discrete case) DTC

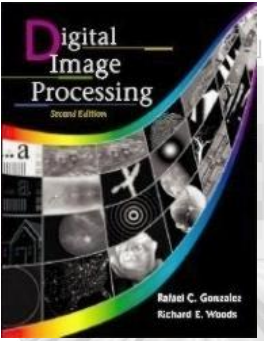
$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-j2\pi ux/M} \quad \text{for } u = 0,1,2,\ldots,M-1$$

  - Inverse Fourier transform:

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$f(x) = \sum_{u=0}^{M-1} F(u)e^{j2\pi ux/M} \quad \text{for } x = 0,1,2,\ldots,M-1$$

© 2002 R. C. Gonzalez & R. E. Woods



Introduction to the Fourier Transform and the Frequency Domain

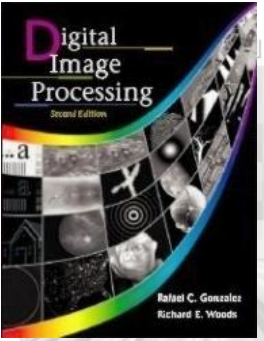
- Since $e^{j\theta} = \cos\theta + j\sin\theta$ and the fact $\cos(-\theta) = \cos\theta$ then discrete Fourier transform can be redefined

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)[\cos 2\pi ux/M - j\sin 2\pi ux/M]$$

$$\text{for } u = 0,1,2,\ldots,M-1$$

  - Frequency (time) domain: the domain (values of u) over which the values of $F(u)$ range; because u determines the frequency of the components of the transform.
  - Frequency (time) component: each of the M terms of $F(u)$.

© 2002 R. C. Gonzalez & R. E. Woods



Introduction to the Fourier Transform and the Frequency Domain

- $F(u)$ can be expressed in polar coordinates:

$$F(u) = |F(u)|e^{j\phi(u)}$$

Value

مسؤول عن قيمة ال Pixel

$$\text{where } |F(u)| = \left[R^2(u) + I^2(u)\right]^{\frac{1}{2}} \quad \text{(magnitude or spectrum)}$$

location

مسؤول عن مكان وجود ال Pixel

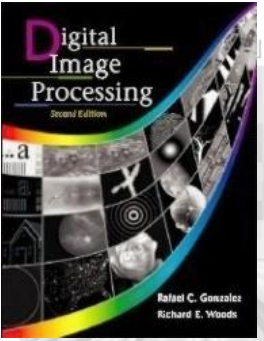
$$\phi(u) = \tan^{-1}\left[\frac{I(u)}{R(u)}\right] \quad \text{(phase angle or phase spectrum)}$$

  - $R(u)$: the real part of $F(u)$
  - $I(u)$: the imaginary part of $F(u)$

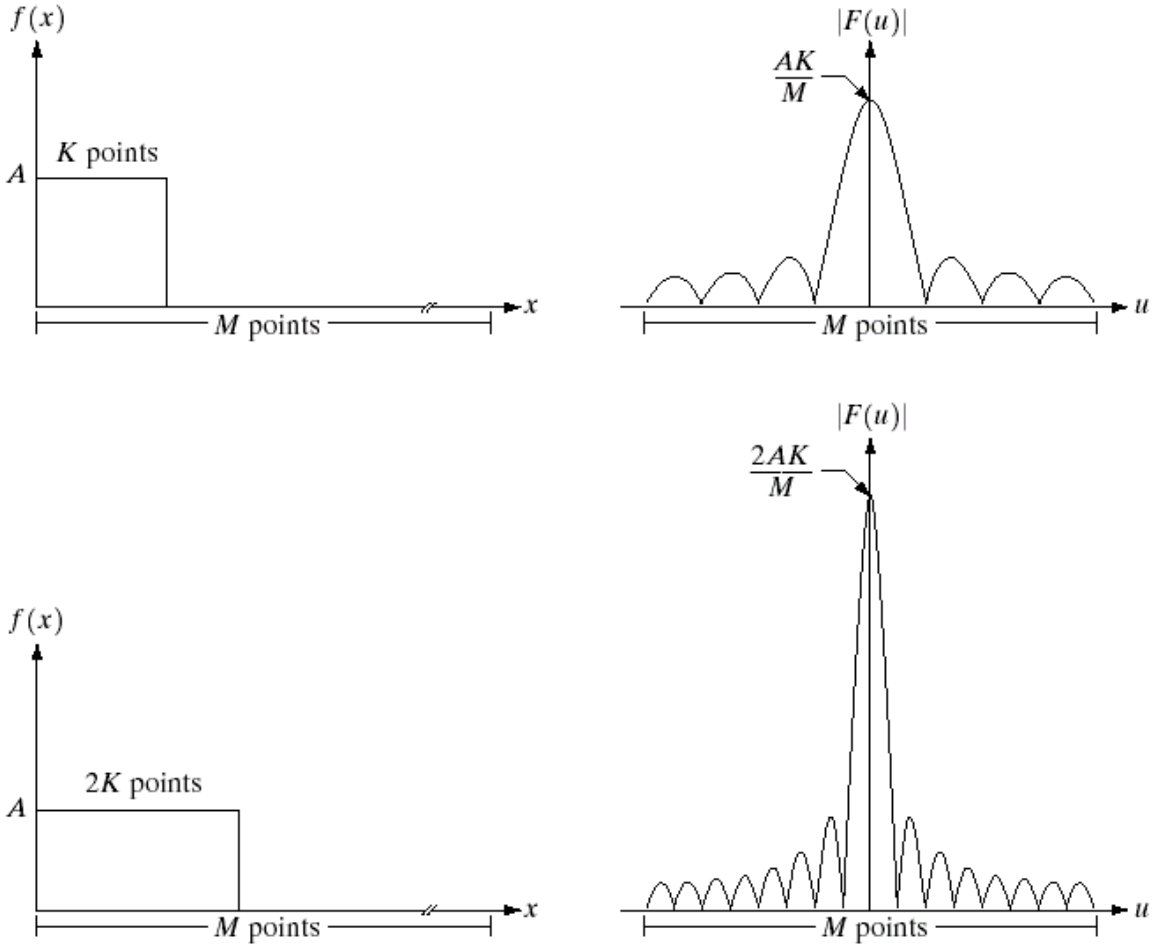
- Power spectrum:

$$P(u) = |F(u)|^2 = R^2(u) + I^2(u)$$

© 2002 R. C. Gonzalez & R. E. Woods



The One-Dimensional Fourier Transform Example



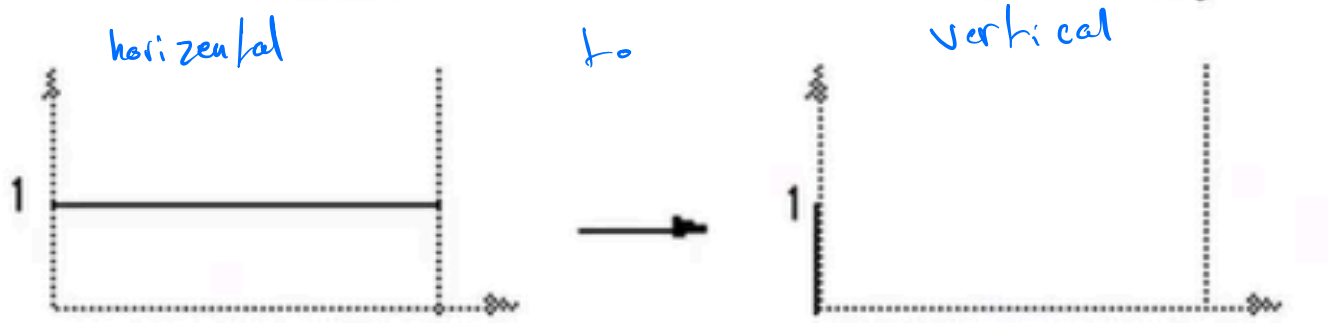
a b
c d

FIGURE 4.2 (a) A discrete function of M points, and (b) its Fourier spectrum. (c) A discrete function with twice the number of nonzero points, and (d) its Fourier spectrum.

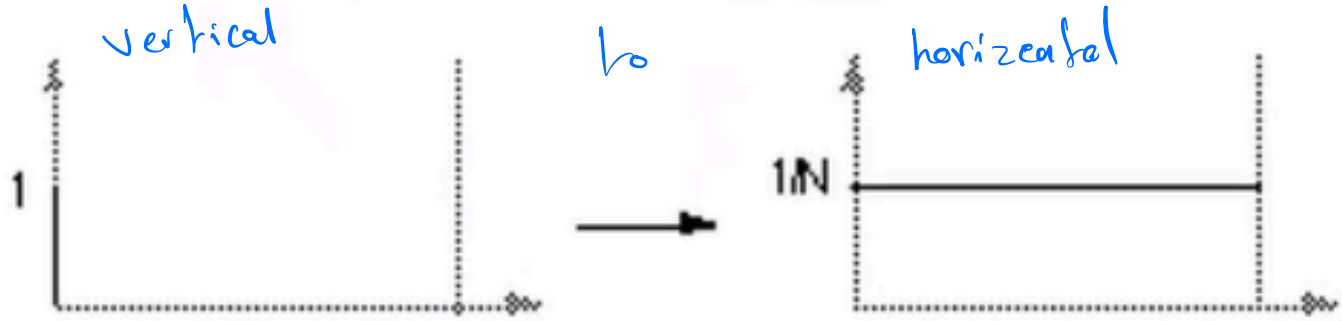
© 2002 R. C. Gonzalez & R. E. Woods

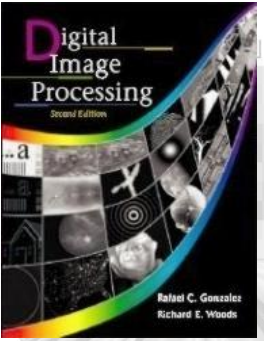
The One-Dimensional Fourier Transform
Some Examples

- The transform of a constant function is a DC value only.



- The transform of a delta function is a constant.





Introduction to the Fourier Transform and the Frequency Domain

- The two-dimensional Fourier transform and its inverse
  - Fourier transform (discrete case) DTC

$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$$

$$\text{for } u = 0,1,2,\ldots,M-1, v = 0,1,2,\ldots,N-1$$

  - Inverse Fourier transform:

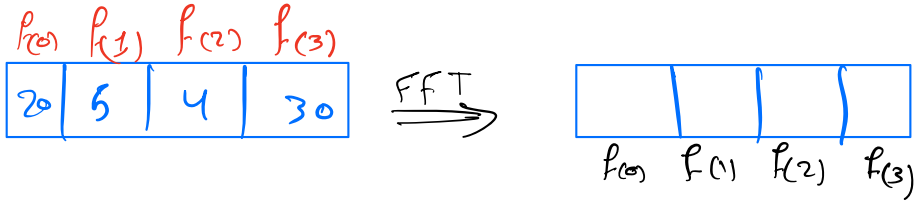
$$f(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)e^{j2\pi(ux/M+vy/N)}$$

$$\text{for } x = 0,1,2,\ldots,M-1, y = 0,1,2,\ldots,N-1$$

- u, v: the transform or frequency variables
- x, y: the spatial or image variables

© 2002 R. C. Gonzalez & R. E. Woods





$$F(1) = \frac{1}{4}\sum_{x=0}^{3} f(x)\, e^{-j2\pi ux/M}$$

$$= \frac{1}{4}\left[20 * e^{-j2\pi * 0/4} + 5 * e^{-j2\pi * 1/4} + 4 * e^{-j2\pi * 2/4} + 30 * e^{-j2\pi * 3/4}\right]$$

$e^{-j\pi} = \cos\pi - j\sin\pi$

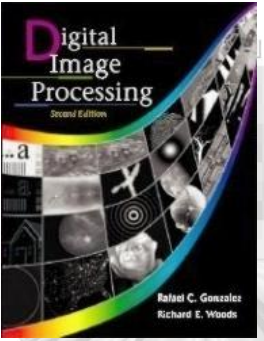
$-j\frac{3\pi}{2} = \cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2}$

$e^{-j\frac{\pi}{2}} = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2}$

$$F(u) = \frac{1}{M}\sum_{x=0}^{M-1} f(x)e^{-j2\pi ux/M}$$

$\cos(-\theta) = \cos\theta$

$\sin(-\theta) = -\sin\theta$



Introduction to the Fourier Transform and the Frequency Domain

- We define the Fourier spectrum, phase anble, and power spectrum as follows:

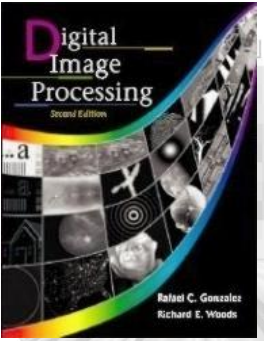
$$|F(u,v)| = \left[R^2(u,v) + I^2(u,v)\right]^{\frac{1}{2}} \quad \text{(spectrum)}$$

$$\phi(u,v) = \tan^{-1}\left[\frac{I(u,v)}{R(u,v)}\right] \quad \text{(phase angle)}$$

$$P(u,v) = |F(u,v)|^2 = R^2(u,v) + I^2(u,v) \quad \text{(power spectrum)}$$

  - $R(u,v)$: the real part of $F(u,v)$
  - $I(u,v)$: the imaginary part of $F(u,v)$

© 2002 R. C. Gonzalez & R. E. Woods



Introduction to the Fourier Transform and the Frequency Domain

- Some properties of Fourier transform:

$$\Im\left[f(x,y)(-1)^{x+y}\right] = F\left(u-\frac{M}{2},v-\frac{N}{2}\right) \quad \text{(shift)}$$

$$F(0,0) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y) \qquad \text{(average)}$$

$$F(u,v) = F^*(-u,-v) \qquad \text{(conujgate symmetric)}$$

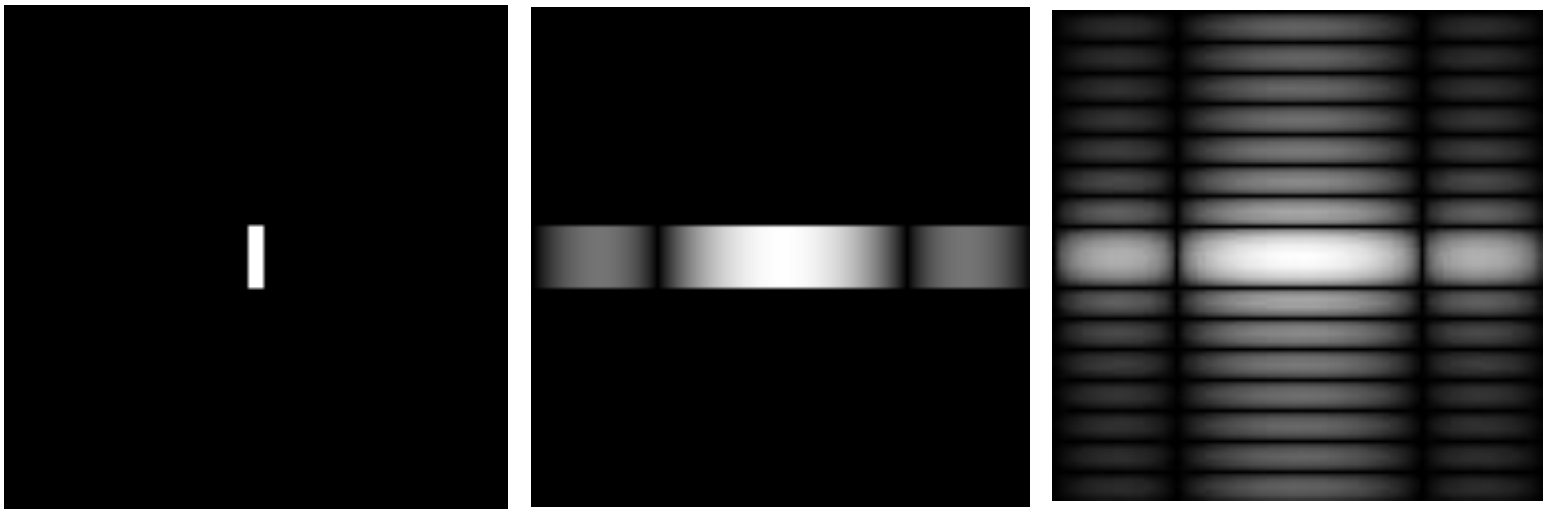
$$\left|F(u,v)\right| = \left|F(-u,-v)\right| \qquad \text{(symmetric)}$$

© 2002 R. C. Gonzalez & R. E. Woods

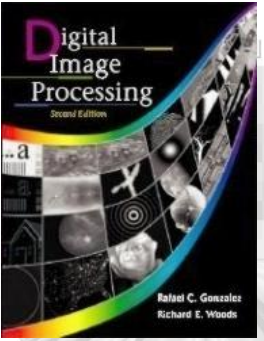
The Two-Dimensional DFT and Its Inverse

The 2D DFT **F(u,v)** can be obtained by

1. taking the 1D DFT of every row of image **f(x,y)**, **F(u,y)**,
2. taking the 1D DFT of every column of **F(u,y)**



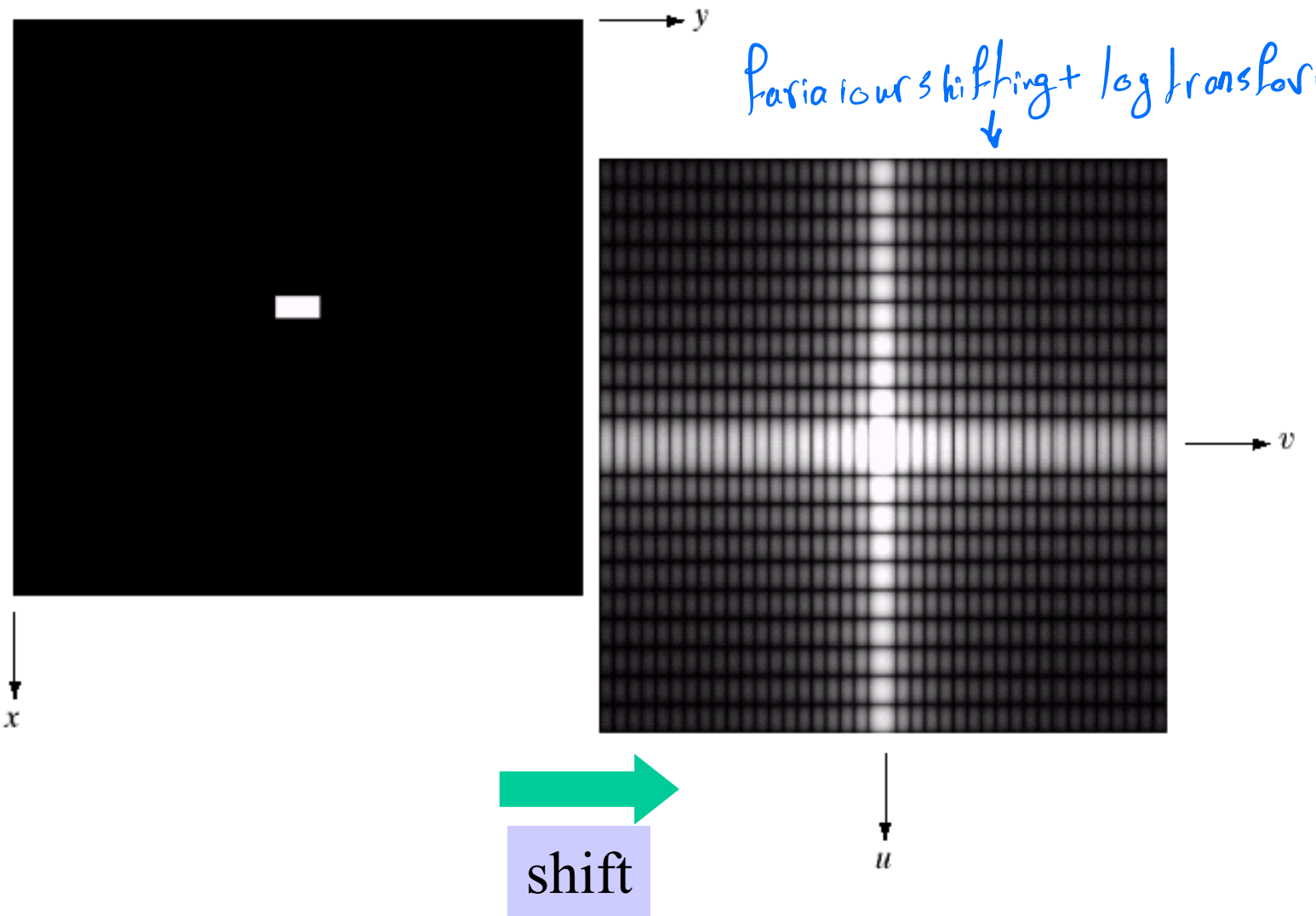
(a)f(x,y) (b)F(u,y) (c)F(u,v)



The Two-Dimensional DFT and Its Inverse

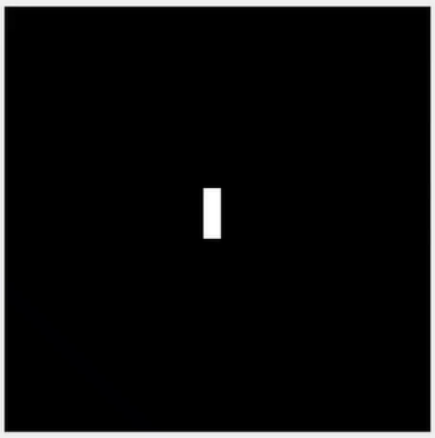
a b

FIGURE 4.3
(a) Image of a 20 × 40 white rectangle on a black background of size 512 × 512 pixels.
(b) Centered Fourier spectrum shown after application of the log transformation given in Eq. (3.2-2). Compare with Fig. 4.2.

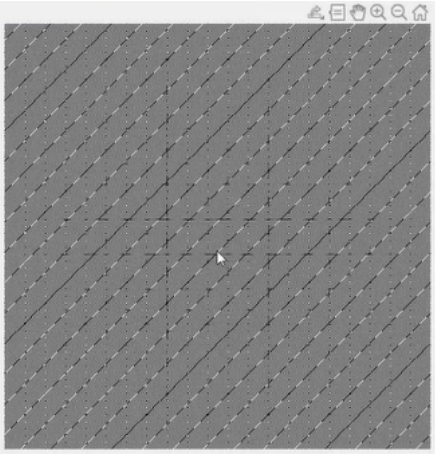


© 2002 R. C. Gonzalez & R. E. Woods

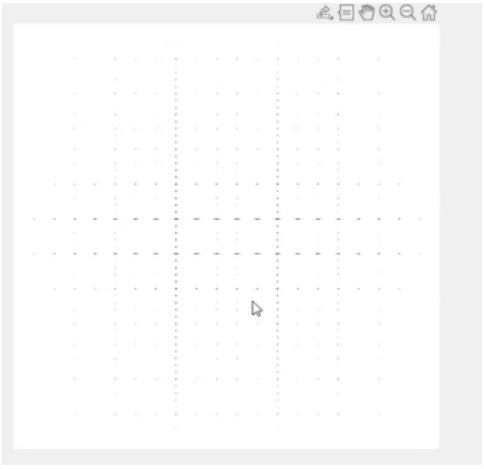
```
i=uint8(zeros(512,512));
i(220:280,240:260)=255;
imshow(i)
```



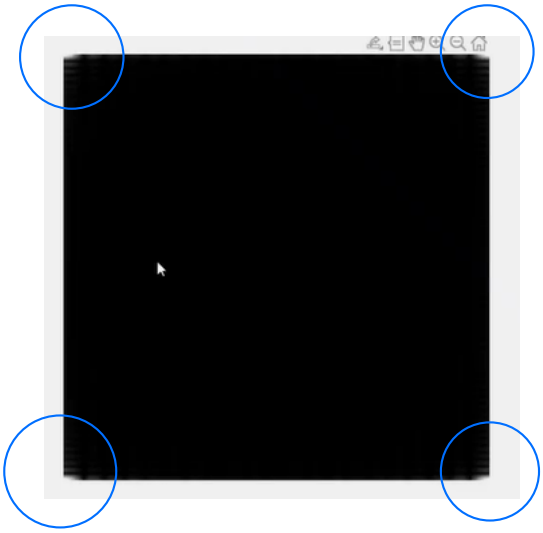
```
i=uint8(zeros(512,512));
i(220:280,240:260)=255;
f=fft2(double(i));
imshow(f)
```



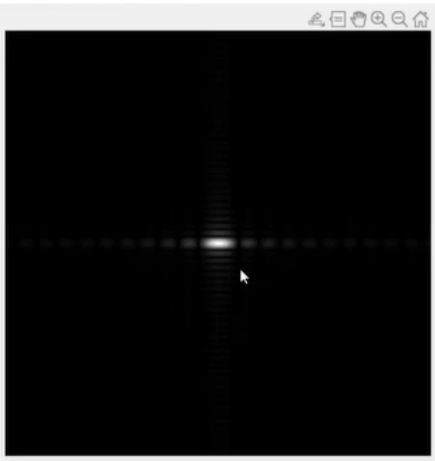
```
i=uint8(zeros(512,512));
i(220:280,240:260)=255;
f=fft2(double(i));
imshow(abs(f))
```



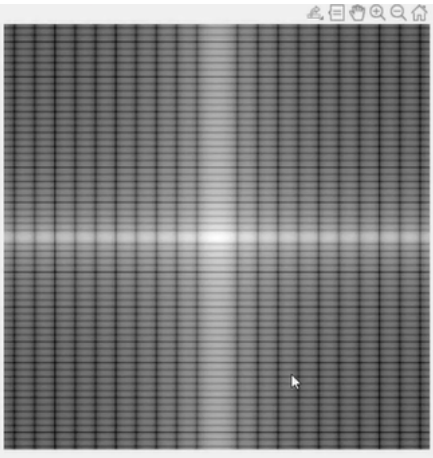
```
i=uint8(zeros(512,512));
i(220:280,240:260)=255;
f=fft2(double(i));
imshow(abs(f),[])
```



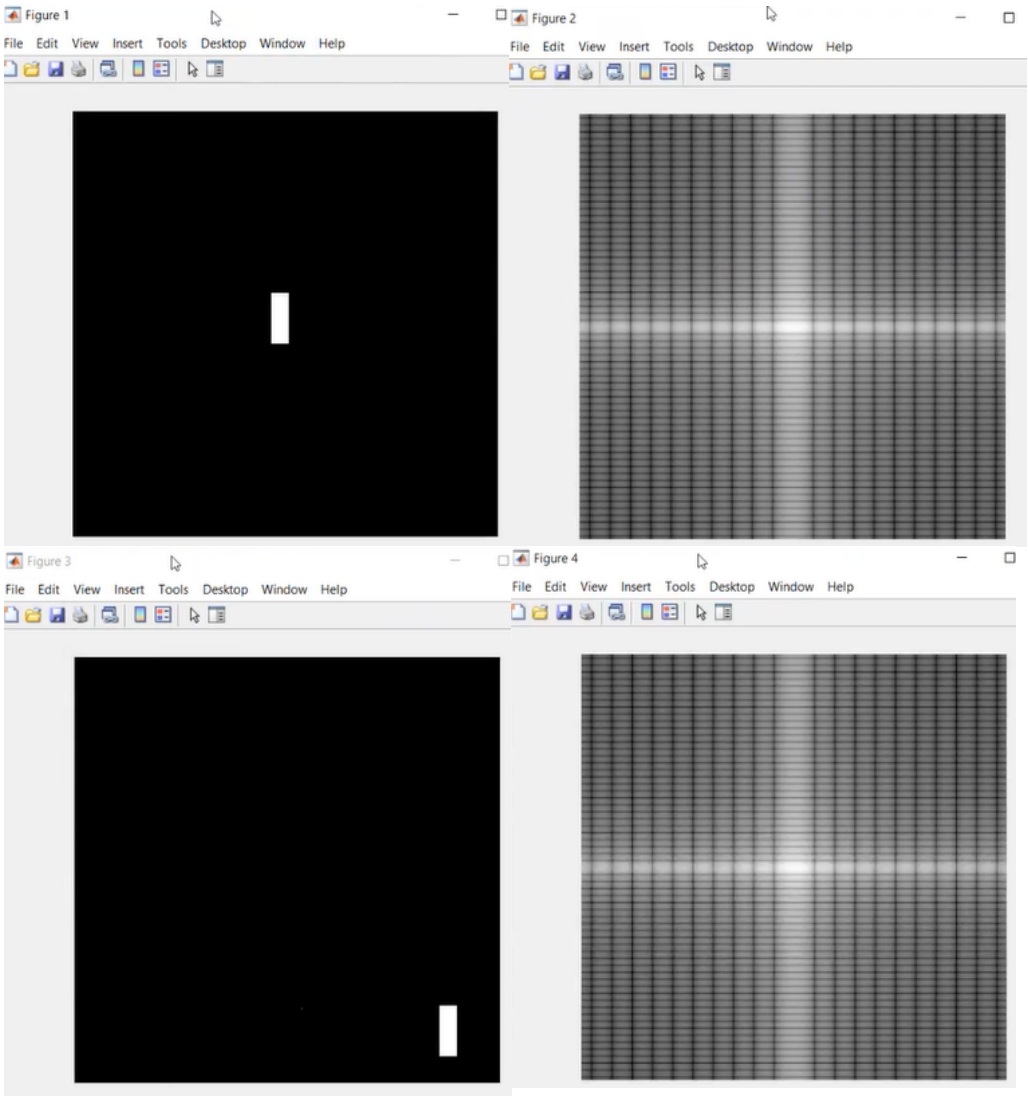
```
i=uint8(zeros(512,512));
i(220:280,240:260)=255;
f=fft2(double(i));
fsh=fftshift(f);
imshow(abs(fsh),[])
```



```
i=uint8(zeros(512,512));
i(220:280,240:260)=255;
f=fft2(double(i));
fsh=fftshift(f);
flog=log(abs(fsh)+1);
imshow(i);figure;imshow(flog,[])
```

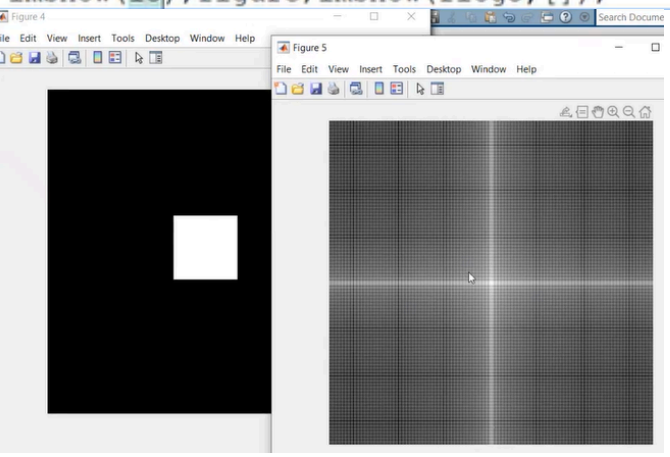


```
i=uint8(zeros(512,512));
i(220:280,240:260)=255;
f=fft2(double(i));
fsh=fftshift(f);
flog=log(abs(fsh)+1);
imshow(i);figure;imshow(flog,[]);
%%%%%after translation%%%%%%
i2=uint8(zeros(512,512));
i2(420:480,440:460)=255;
f2=fft2(double(i2));
fsh2=fftshift(f2);
flog2=log(abs(fsh2)+1);
figure; imshow(i2);figure;imshow(flog2,[]);
```



هذا اثبات انه في حال تغير مكان المستطيل لا تتغير Fourier filter لانه يعتمد على قيم ال pixels مش مكانها.

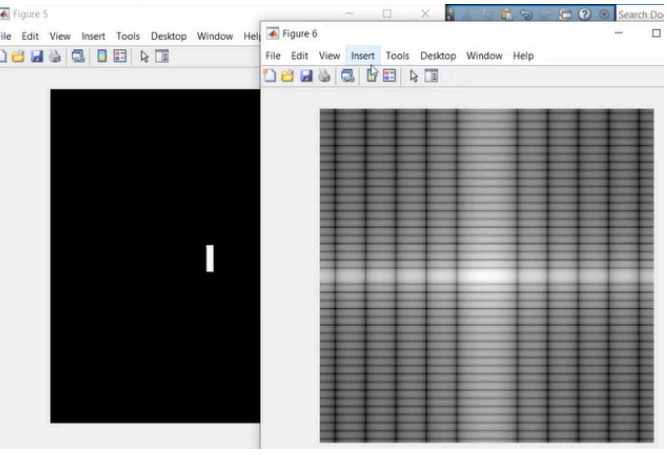
```
%%%%%after translation%%%%%%
i2=uint8(zeros(512,512));
i2(420:480,440:460)=255;
f2=fft2(double(i2));
fsh2=fftshift(f2);
flog2=log(abs(fsh2)+1);
figure; imshow(i2);figure;imshow(flog2,[]);
%%%%% after scaling%%%%%%%% bigger
i3=uint8(zeros(512,512));
i3(200:300,200:300)=255;
f3=fft2(double(i3));
fsh3=fftshift(f3);
flog3=log(abs(fsh3)+1);
imshow(i3);figure;imshow(flog3,[]);
```



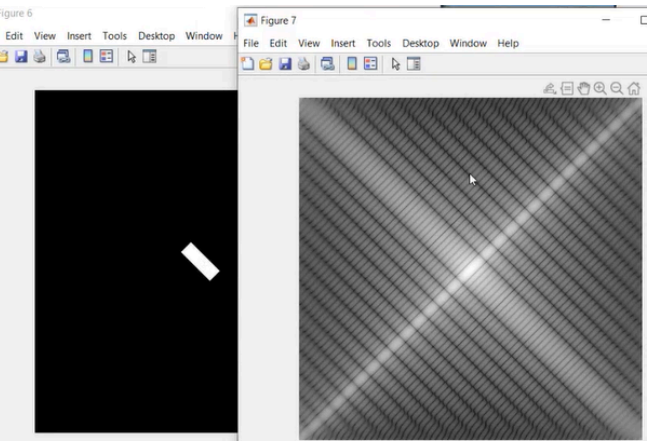
هذا يثبت انه في حال تكبير حجم المستطيل
يزيد عدد Pulses .

*وفي حال تصغير حجم المستطيل
تقل عدد Pulses

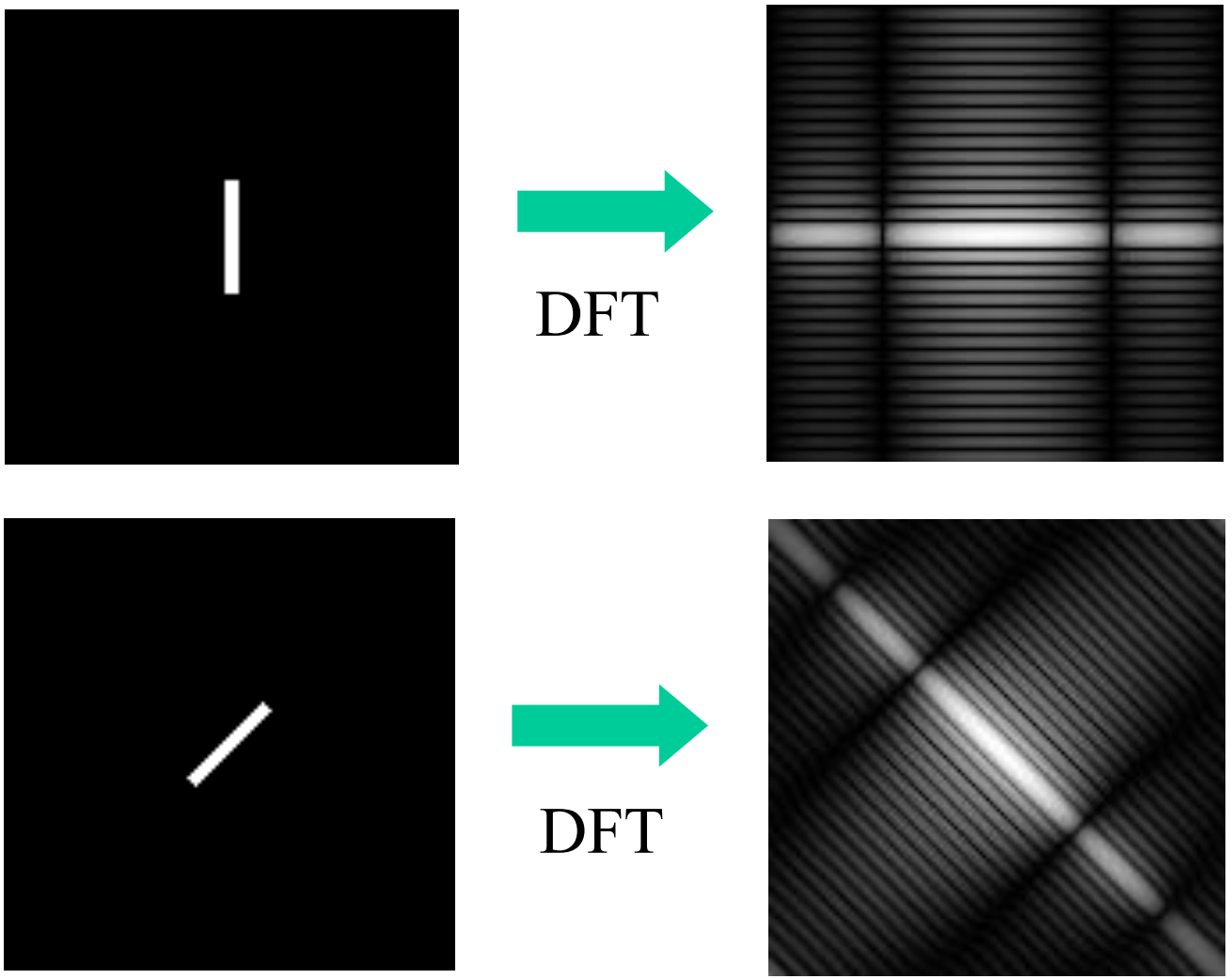
```
%%%%% after scaling (bigger)%%%%%%%%
i3=uint8(zeros(512,512));
i3(200:300,200:300)=255;
f3=fft2(double(i3));
fsh3=fftshift(f3);
flog3=log(abs(fsh3)+1);
imshow(i3);figure;imshow(flog3,[]);
%%%%%%%% after scaling (smaller)%%%%%%%
i4=uint8(zeros(512,512));
i4(240:280,240:250)=255;
f4=fft2(double(i4));
fsh4=fftshift(f4);
flog4=log(abs(fsh4)+1);
imshow(i4);figure;imshow(flog4,[]);
```



```
%%%%%% after rotation %%%%%
i5=imrotate(i,45,'bilinear','crop');
f5=fft2(double(i5));
fsh5=fftshift(f5);
flog5=log(abs(fsh5)+1);
imshow(i5);figure;imshow(flog5,[]);
```

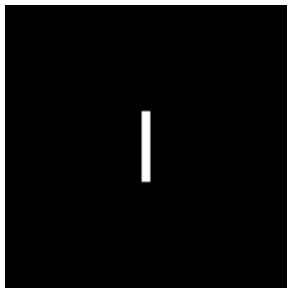


The Property of Two-Dimensional DFT Rotation

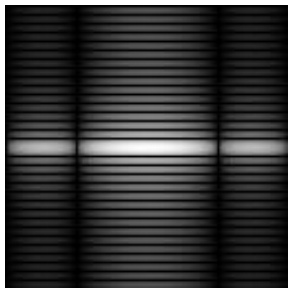


The Property of Two-Dimensional DFT
Linear Combination

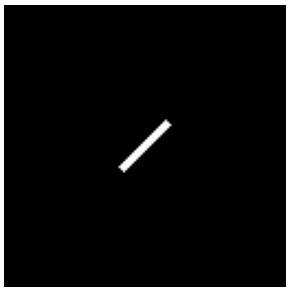
A



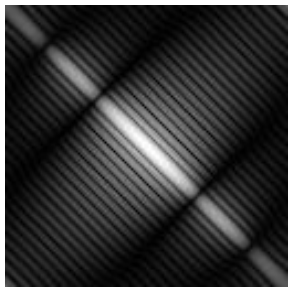
DFT



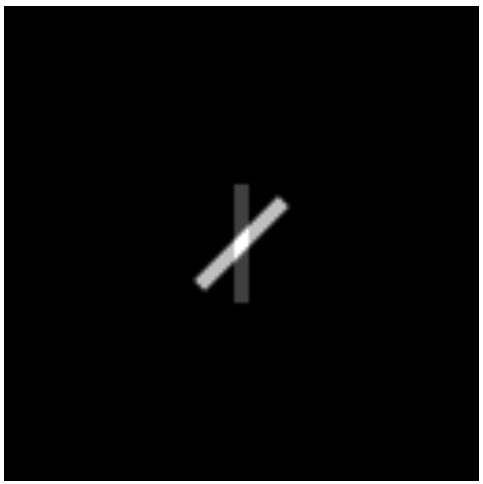
B



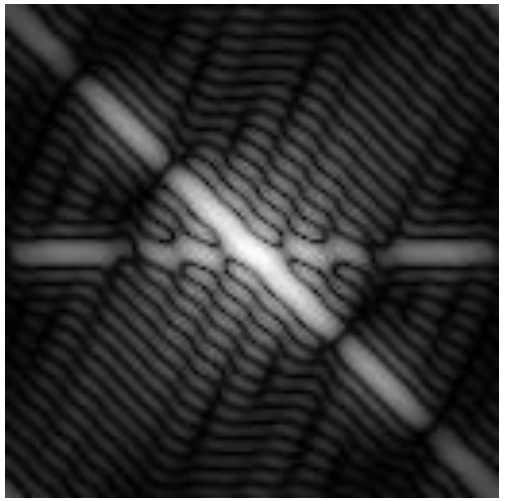
DFT



0.25 * A
+ 0.75 * B

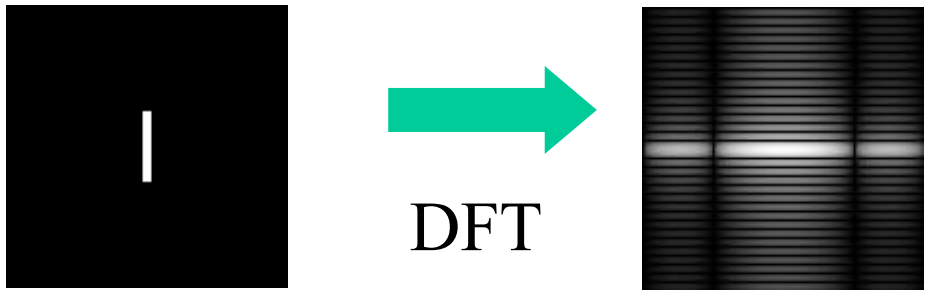


DFT

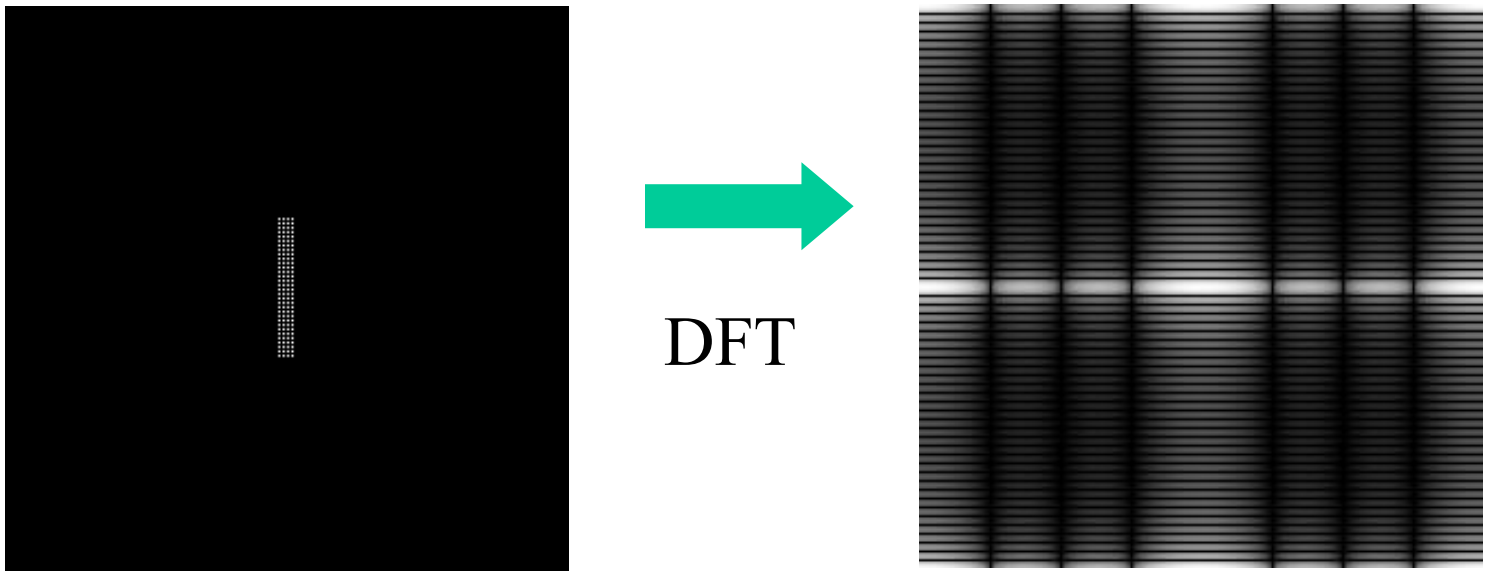


The Property of Two-Dimensional DFT Expansion

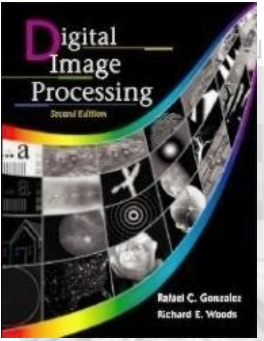
A



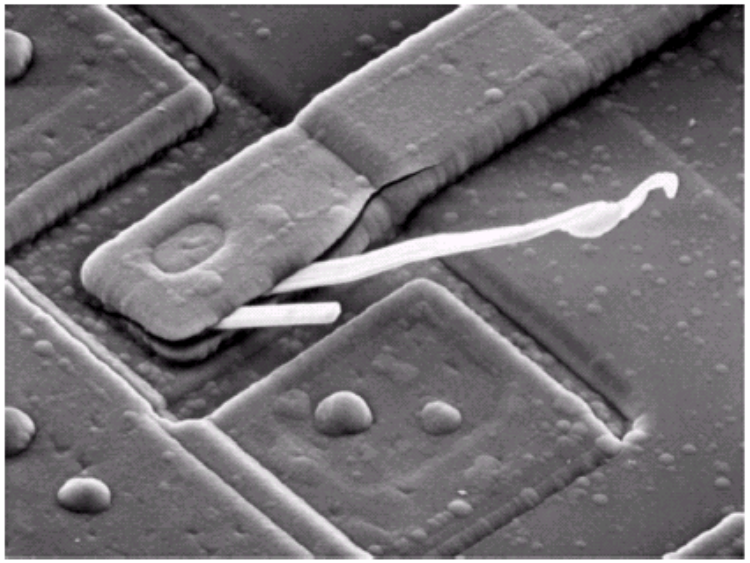
B

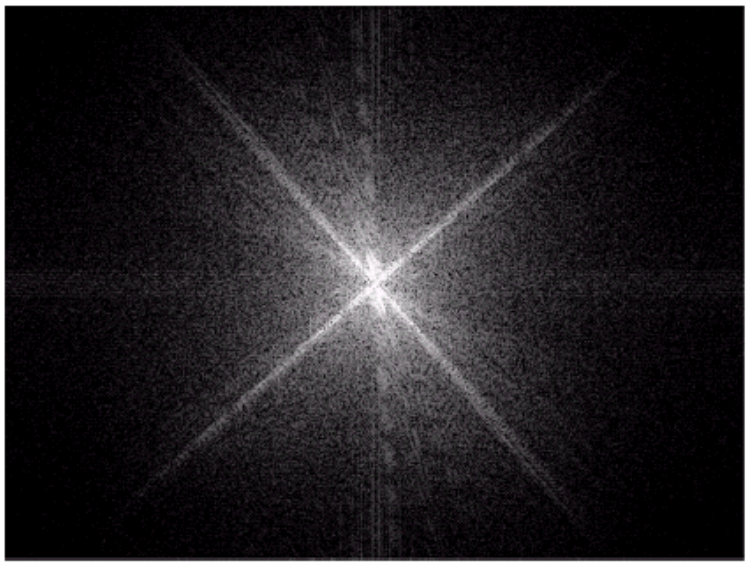


Expanding the original image by a factor of n (n=2), filling the empty new values with zeros, results in the same DFT.



Filtering in the Frequency Domain

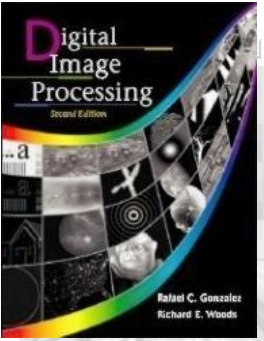




a
b

FIGURE 4.4
(a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

© 2002 R. C. Gonzalez & R. E. Woods



Basics of Filtering in the Frequency Domain

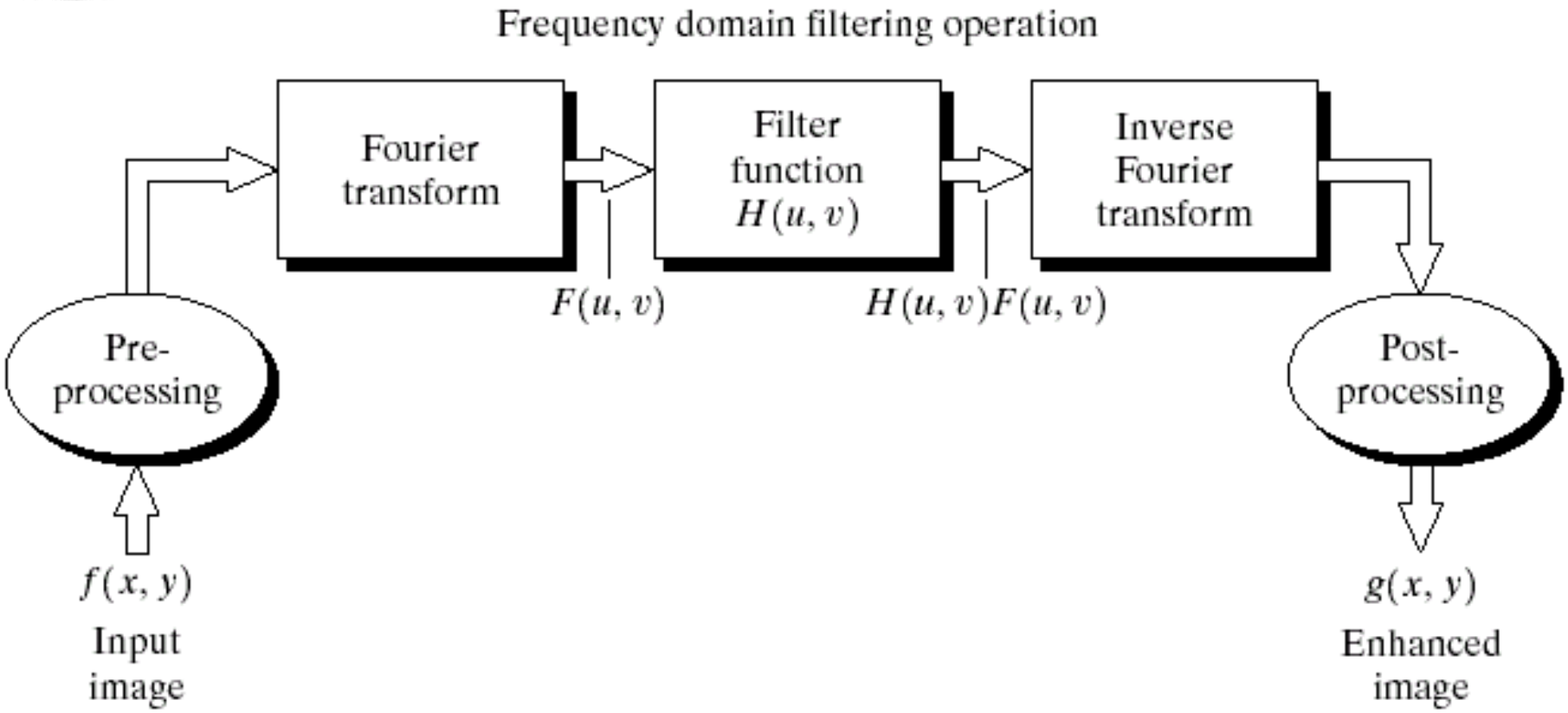
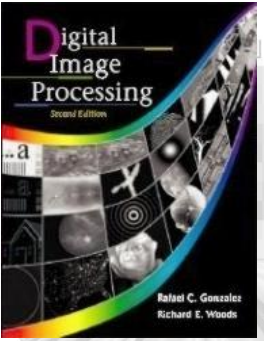


FIGURE 4.5 Basic steps for filtering in the frequency domain.

© 2002 R. C. Gonzalez & R. E. Woods



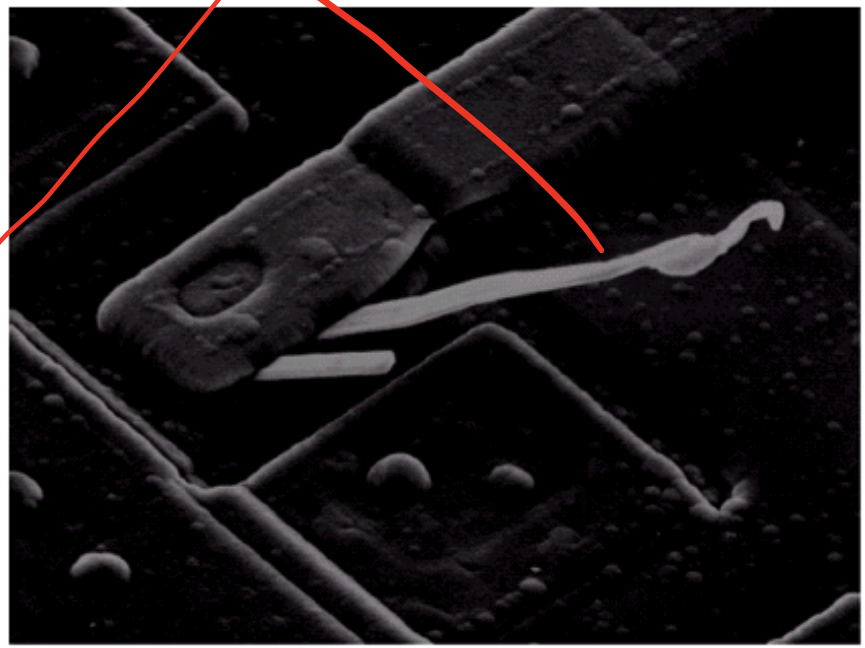
Some Basic Filters and Their Functions

- Multiply all values of $F(u,v)$ by the filter function (notch filter):

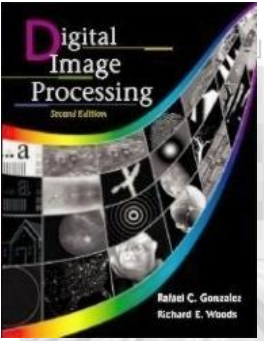
$$H(u,v) = \begin{cases} 0 & \text{if } (u,v) = (M/2, N/2) \\ 1 & \text{otherwise.} \end{cases}$$

  - All this filter would do is set $F(0,0)$ to zero (force the average value of an image to zero) and leave all frequency components of the Fourier transform untouched.

FIGURE 4.6
Result of filtering the image in Fig. 4.4(a) with a notch filter that set to 0 the $F(0, 0)$ term in the Fourier transform.



© 2002 R. C. Gonzalez & R. E. Woods



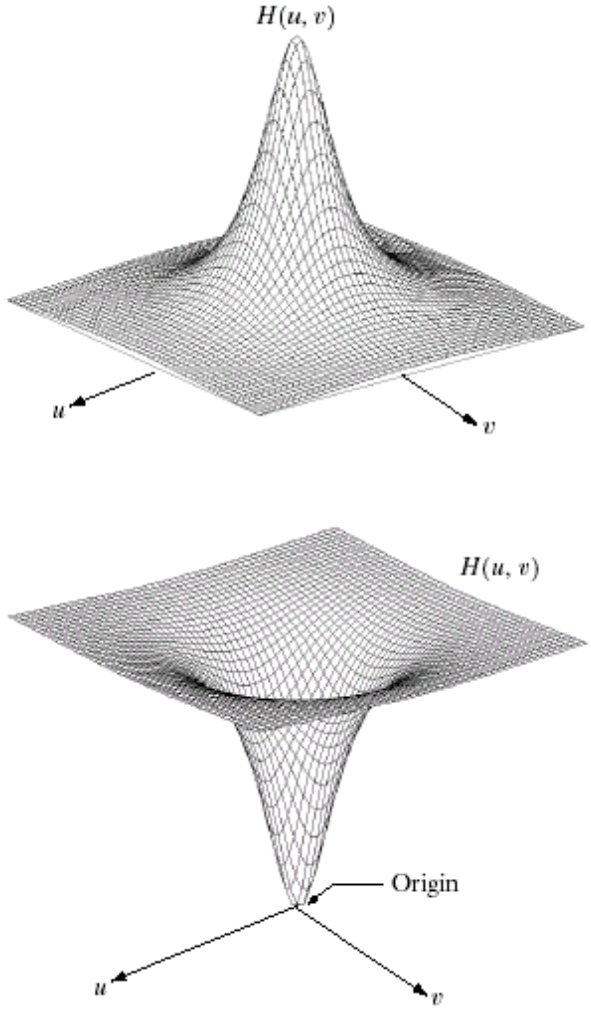
Some Basic Filters and Their Functions

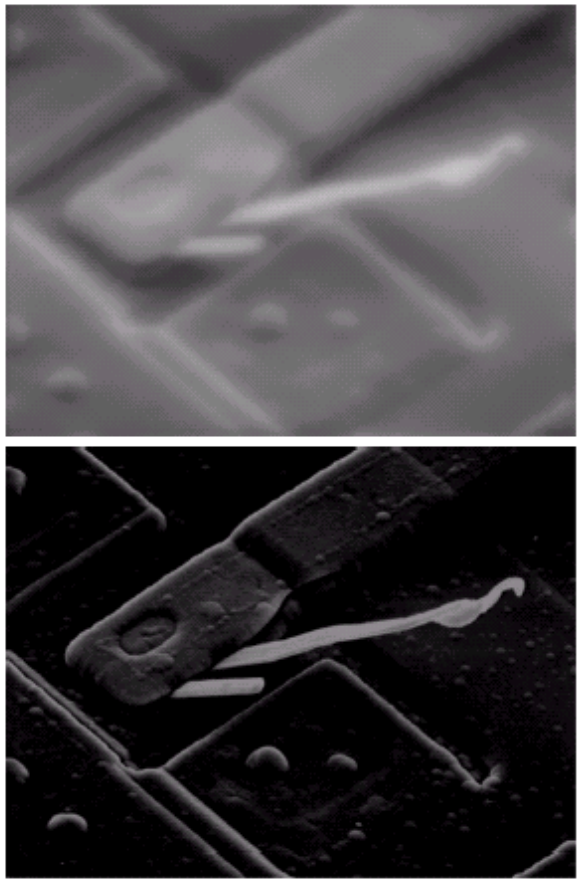
Lowpass filter

Background
Texture

Highpass filter

Edges
Noise

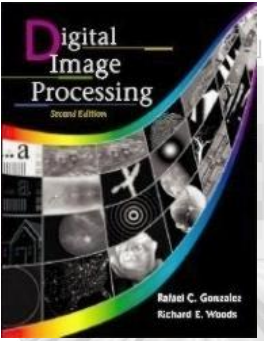




a b
c d

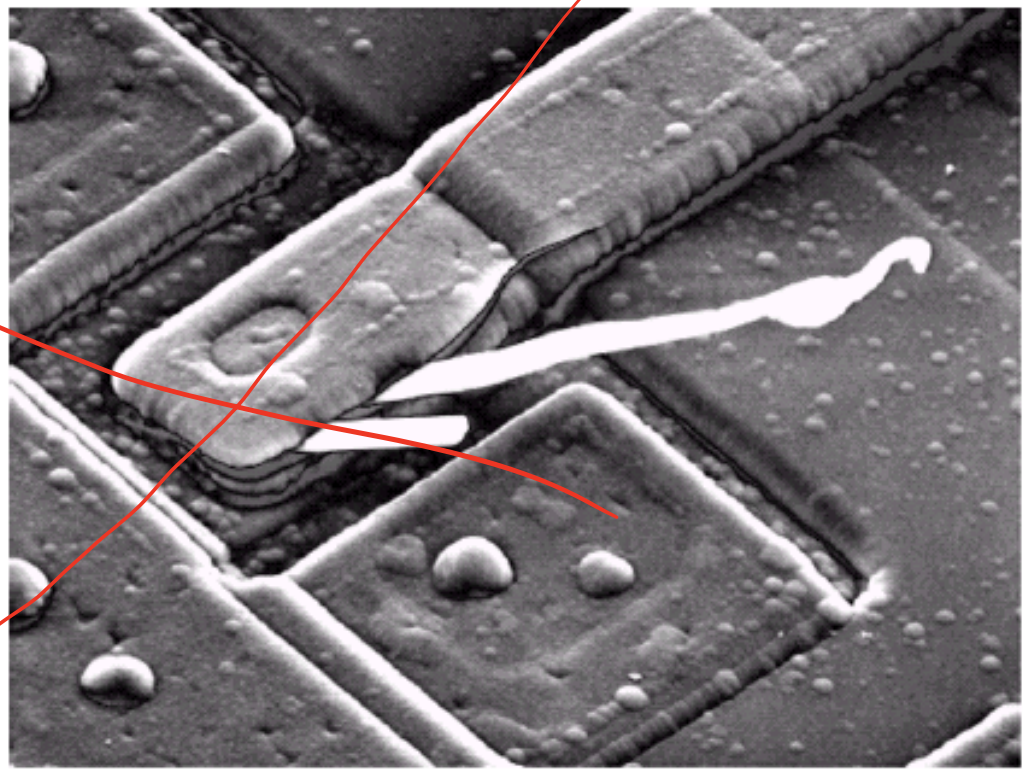
FIGURE 4.7 (a) A two-dimensional lowpass filter function. (b) Result of lowpass filtering the image in Fig. 4.4(a). (c) A two-dimensional highpass filter function. (d) Result of highpass filtering the image in Fig. 4.4(a).

© 2002 R. C. Gonzalez & R. E. Woods

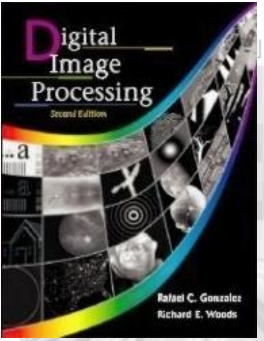


Some Basic Filters and Their Functions

FIGURE 4.8
Result of highpass filtering the image in Fig. 4.4(a) with the filter in Fig. 4.7(c), modified by adding a constant of one-half the filter height to the filter function. Compare with Fig. 4.4(a).



© 2002 R. C. Gonzalez & R. E. Woods



Correspondence between Filtering in the Spatial and Frequency Domain

- Convolution theorem:
  - The discrete convolution of two functions $f(x,y)$ and $h(x,y)$ of size MXN is defined as

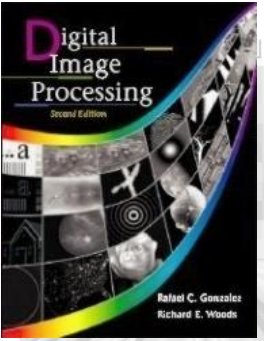
$$f(x,y) * h(x,y) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m,n) h(x-m, y-n)$$

  - Let $F(u,v)$ and $H(u,v)$ denote the Fourier transforms of $f(x,y)$ and $h(x,y)$, then

$$f(x,y) * h(x,y) \Leftrightarrow F(u,v)H(u,v) \qquad \text{Eq. (4.2-31)}$$

$$f(x,y)h(x,y) \Leftrightarrow F(u,v) * H(u,v) \qquad \text{Eq. (4.2-32)}$$

© 2002 R. C. Gonzalez & R. E. Woods



Smoothing Frequency-Domain Filters

- The basic model for filtering in the frequency domain

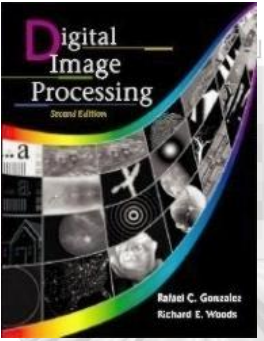
$$G(u,v) = H(u,v)F(u,v)$$

where $F(u,v)$: the Fourier transform of the image to be smoothed

$H(u,v)$: a filter transfer function

- Smoothing is fundamentally a lowpass operation in the frequency domain.
- There are several standard forms of lowpass filters (LPF).
  1. Ideal lowpass filter
  2. Butterworth lowpass filter
  3. Gaussian lowpass filter

© 2002 R. C. Gonzalez & R. E. Woods



Ideal Lowpass Filters (ILPFs)

- The simplest lowpass filter is a filter that "cuts off" all high-frequency components of the Fourier transform that are at a distance greater than a specified distance D_0 from the origin of the transform.
- The transfer function of an ideal lowpass filter

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

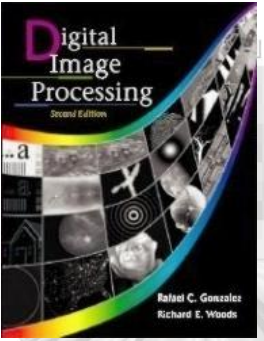
where $D(u,v)$: the distance from point (u,v) to the center of ther frequency rectangle

$$D(u,v) = \left[(u - M/2)^2 + (v - N/2)^2\right]^{\frac{1}{2}}$$

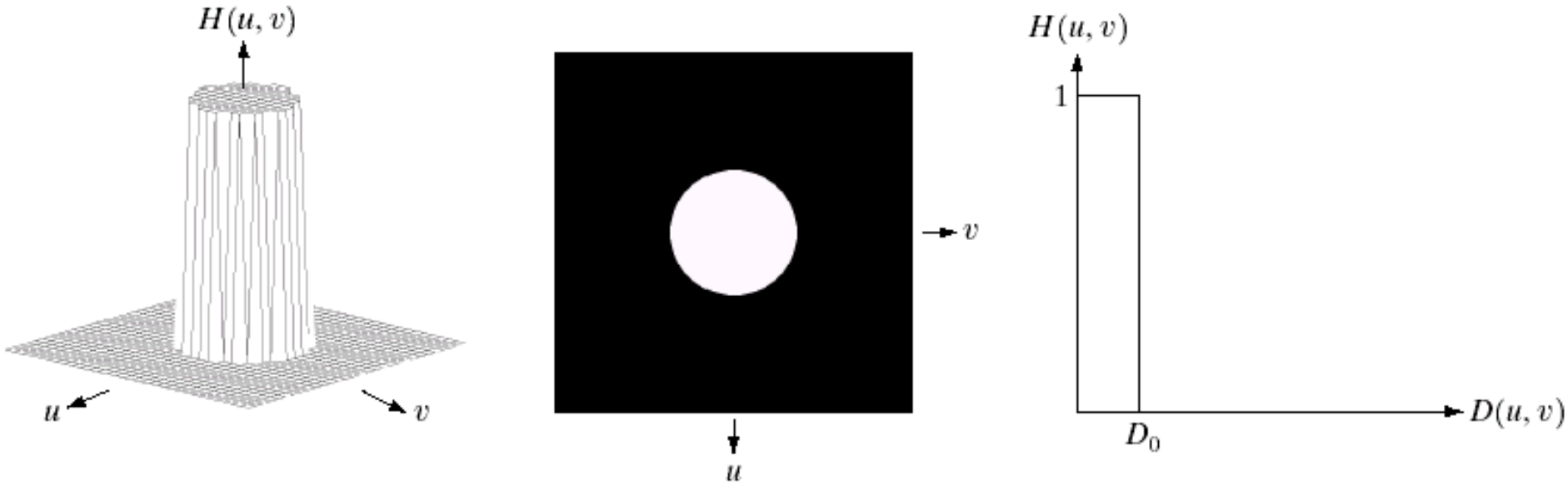
M, N
width, Height

المسافة من الـ Pixel الحالي او جوه المستطيل — نقطة المركز

© 2002 R. C. Gonzalez & R. E. Woods



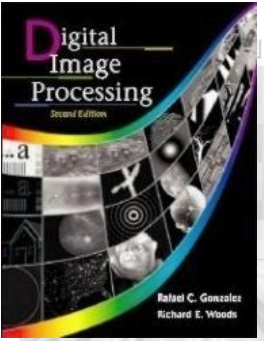
Ideal Lowpass Filters (ILPFs)



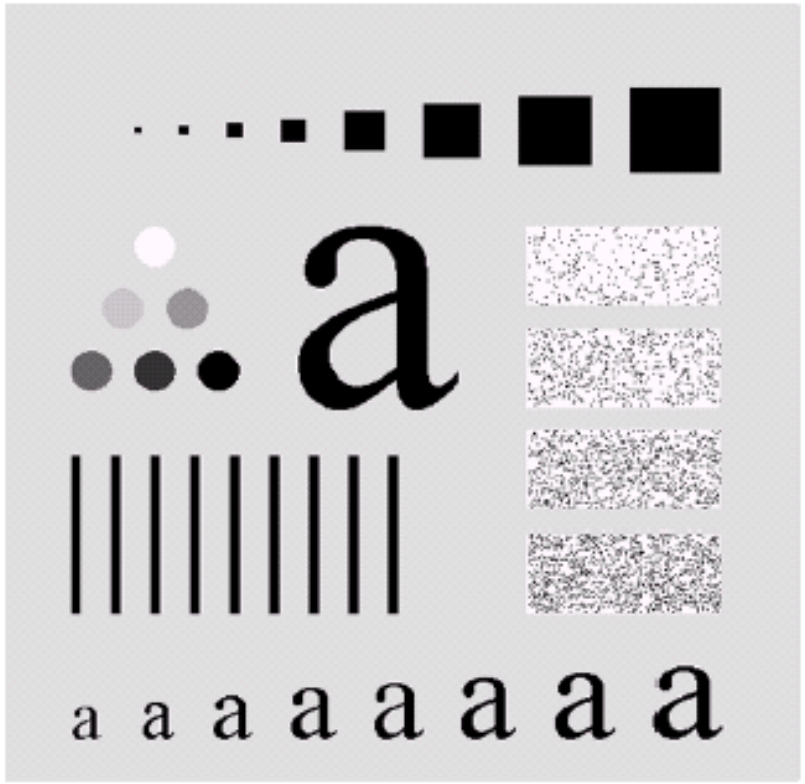
a b c

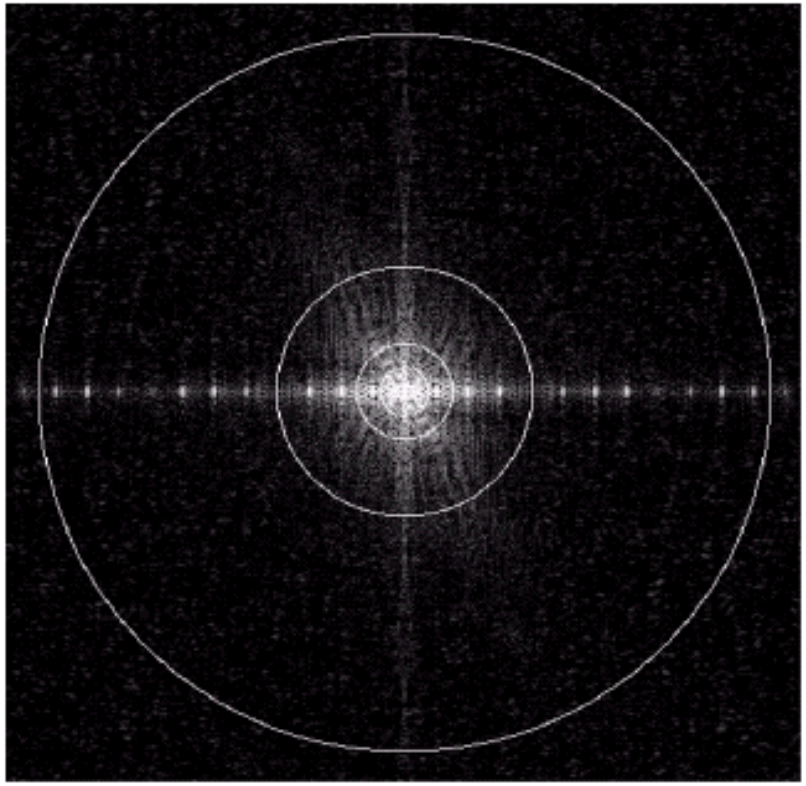
FIGURE 4.10 (a) Perspective plot of an ideal lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

© 2002 R. C. Gonzalez & R. E. Woods



Ideal Lowpass Filters (ILPFs)

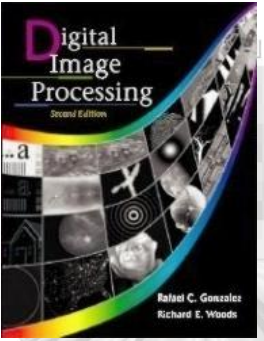




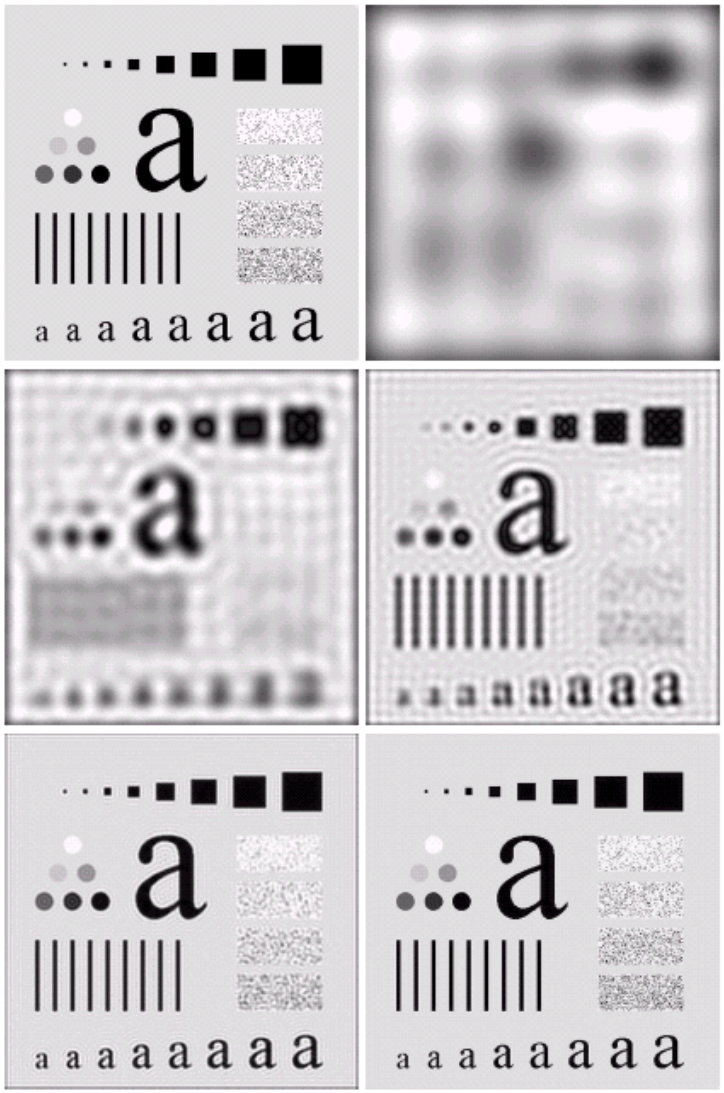
a b

FIGURE 4.11 (a) An image of size 500 × 500 pixels and (b) its Fourier spectrum. The superimposed circles have radii values of 5, 15, 30, 80, and 230, which enclose 92.0, 94.6, 96.4, 98.0, and 99.5% of the image power, respectively.

© 2002 R. C. Gonzalez & R. E. Woods



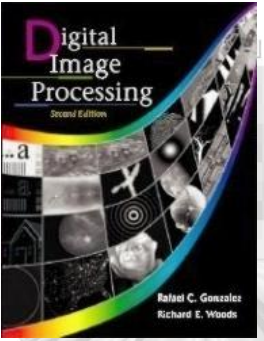
Ideal Lowpass Filters



a b
c d
e f

FIGURE 4.12 (a) Original image. (b)–(f) Results of ideal lowpass filtering with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). The power removed by these filters was 8, 5.4, 3.6, 2, and 0.5% of the total, respectively.

© 2002 R. C. Gonzalez & R. E. Woods



Ideal Lowpass Filters Another Example

Figure 4.13 (a) A frequency-domain ILPF of radius 5. (b) Corresponding spatial filter. (c) Five impulses in the spatial domain, simulating the values of five pixels. (d) Convolution of (b) and (c) in the spatial domain.

$$f(x,y) * h(x,y) \Leftrightarrow F(u,v)H(u,v)$$

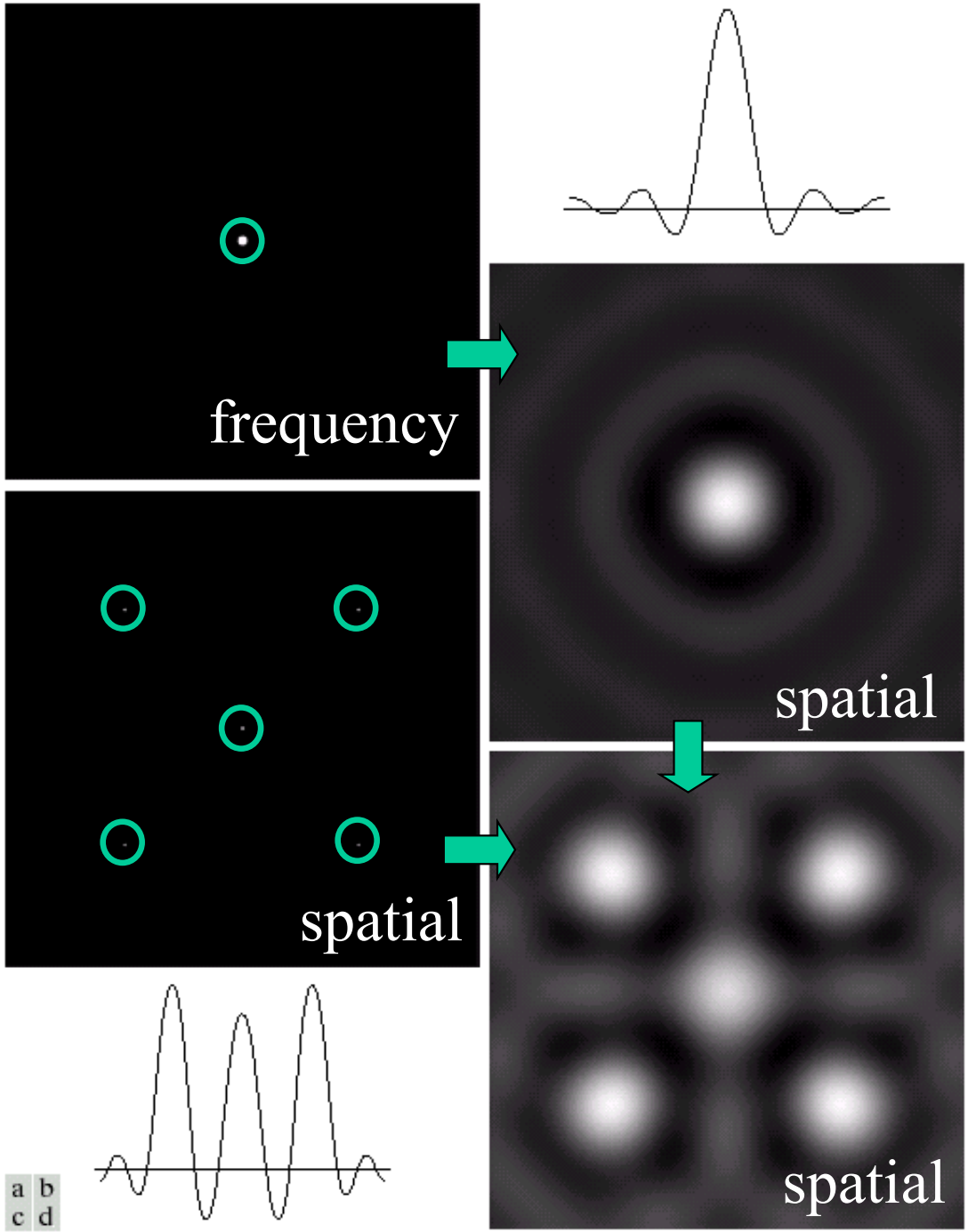
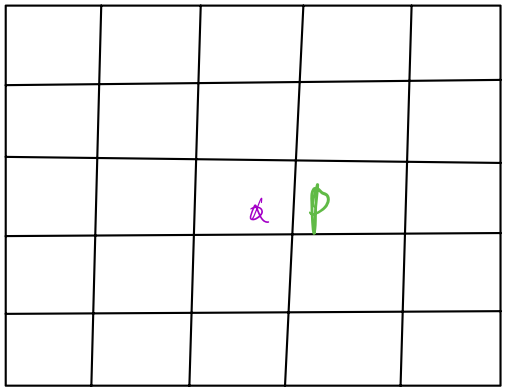
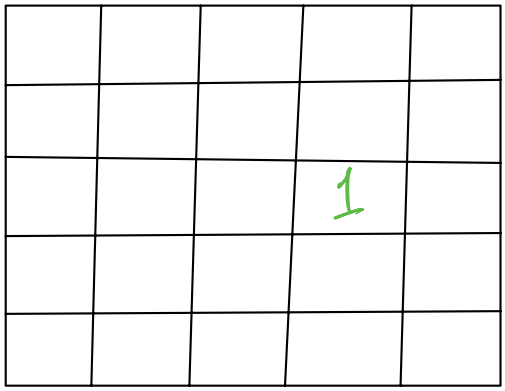


FIGURE 4.13 (a) A frequency-domain ILPF of radius 5. (b) Corresponding spatial filter (note the ringing). (c) Five impulses in the spatial domain, simulating the values of five pixels. (d) Convolution of (b) and (c) in the spatial domain.

© 2002 R. C. Gonzalez & R. E. Woods





center

$M/2 = 3$

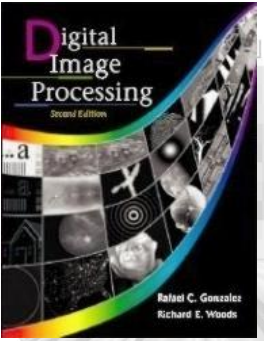
$N/2 = 3$

$D_0 = 3$

$D_p = \text{sqrt}((3-3)^2 + (4-3)^2$

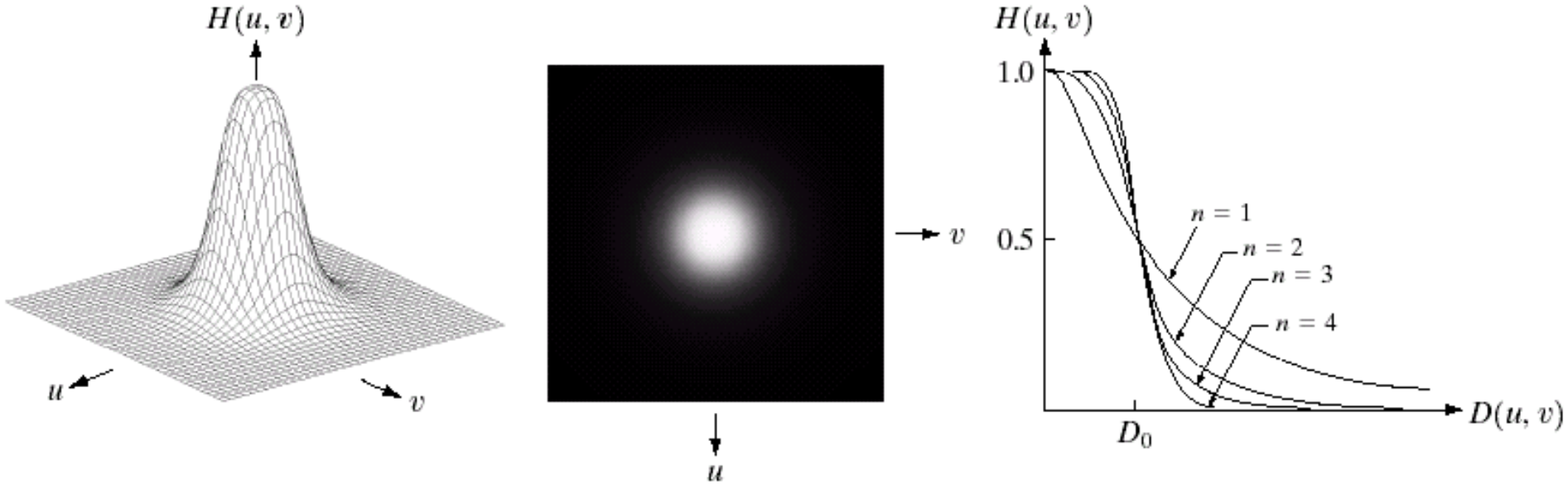
$D_p = 1$

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$



Butterworth Lowpass Filters (BLPFs) With order *n*

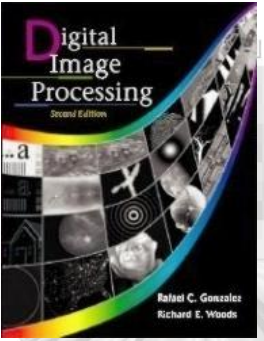
$$H(u,v) = \frac{1}{1+\left[D(u,v)/D_0\right]^{2n}}$$



a b c

FIGURE 4.14 (a) Perspective plot of a Butterworth lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

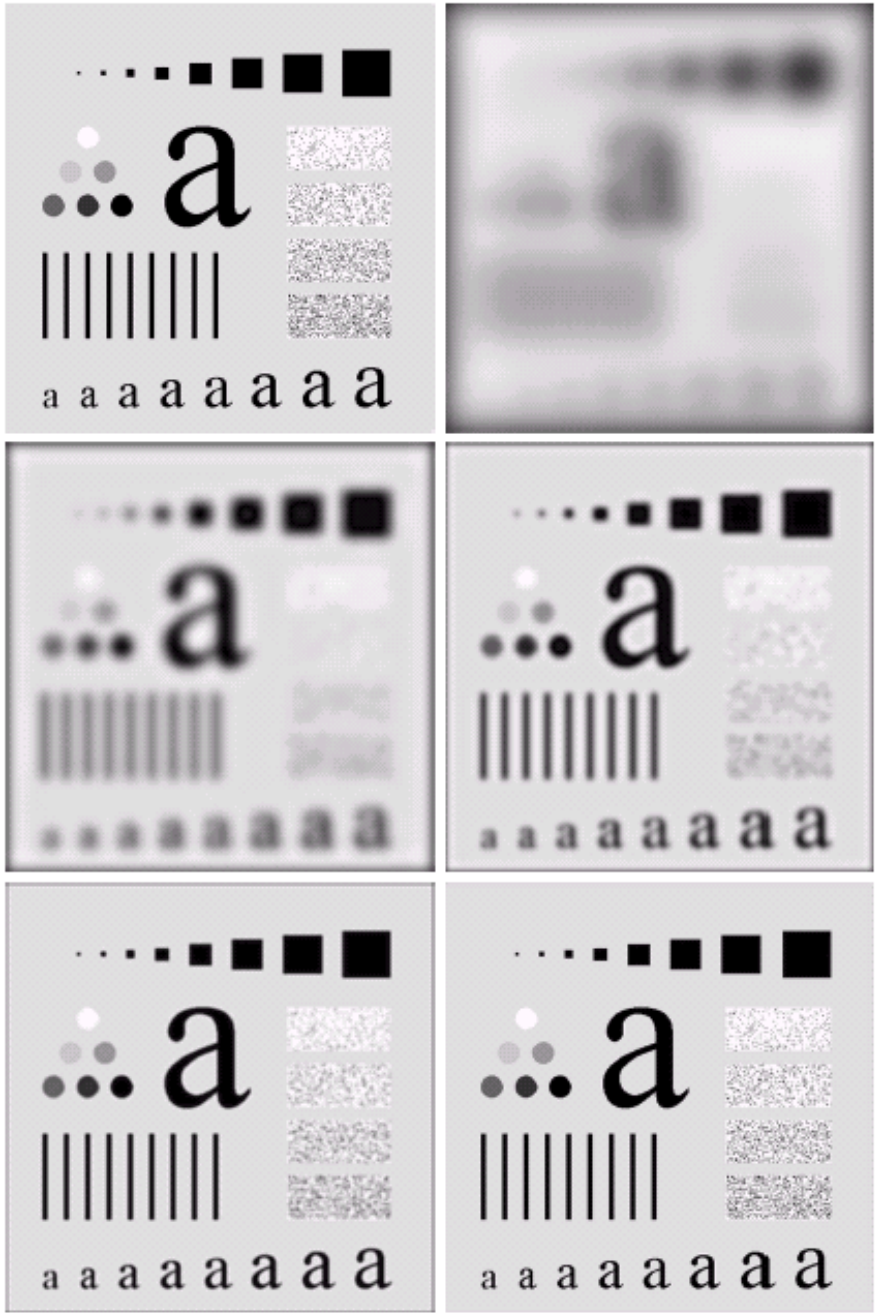
© 2002 R. C. Gonzalez & R. E. Woods



Butterworth Lowpass Filters (BLPFs)

$n=2$

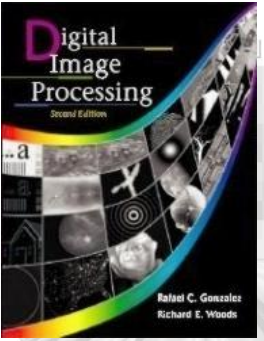
$D_0=5,15,30,80,\text{and } 230$



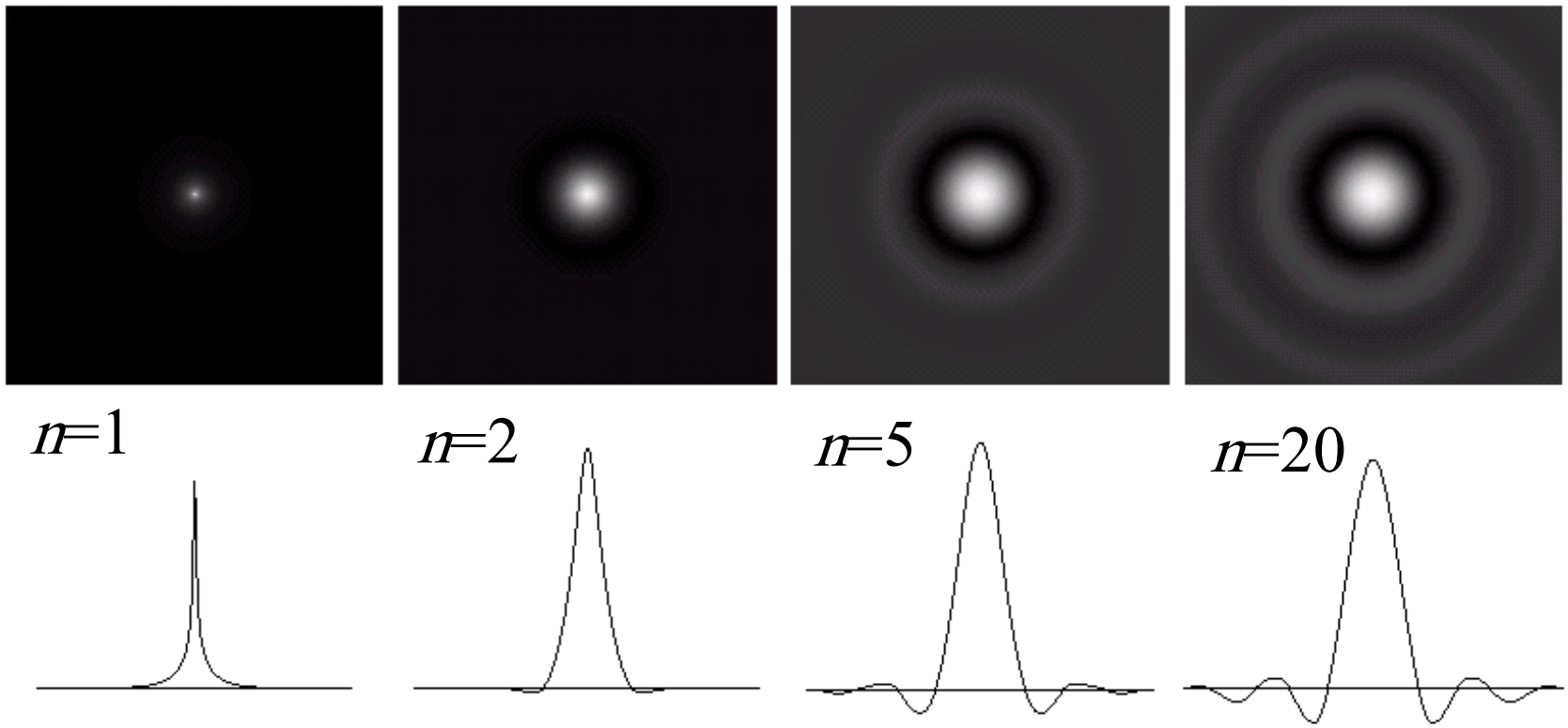
a b
c d
e f

FIGURE 4.15 (a) Original image. (b)–(f) Results of filtering with BLPFs of order 2, with cutoff frequencies at radii of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Fig. 4.12.

© 2002 R. C. Gonzalez & R. E. Woods



Butterworth Lowpass Filters (BLPFs)
Spatial Representation

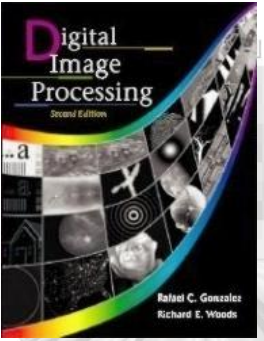


ringing

a b c d

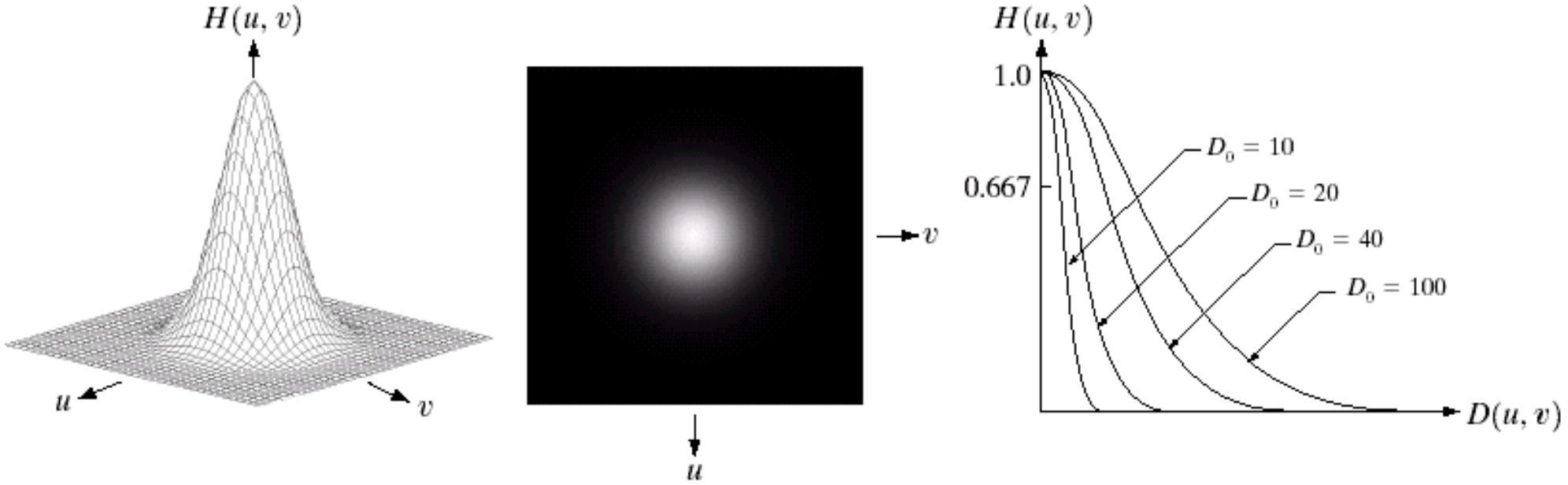
FIGURE 4.16 (a)–(d) Spatial representation of BLPFs of order 1, 2, 5, and 20, and corresponding gray-level profiles through the center of the filters (all filters have a cutoff frequency of 5). Note that ringing increases as a function of filter order.

© 2002 R. C. Gonzalez & R. E. Woods



Gaussian Lowpass Filters (FLPFs)

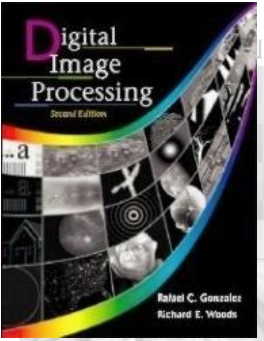
$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$



a b c

FIGURE 4.17 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0.

© 2002 R. C. Gonzalez & R. E. Woods



Gaussian Lowpass Filters (FLPFs)

D_0=5,15,30,80,and 230

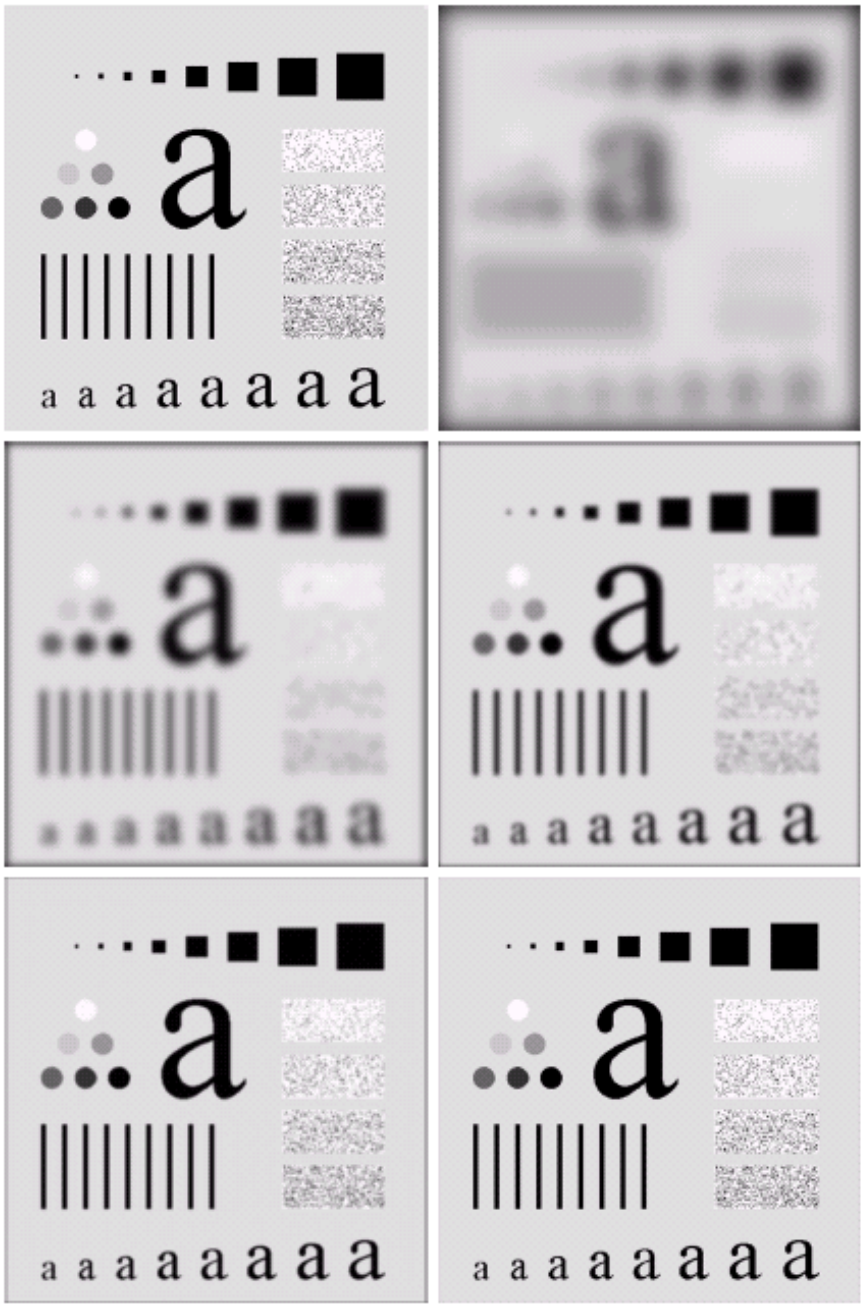
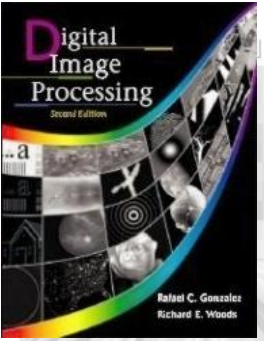


FIGURE 4.18 (a) Original image. (b)–(f) Results of filtering with Gaussian lowpass filters with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Figs. 4.12 and 4.15.

a b
c d
e f

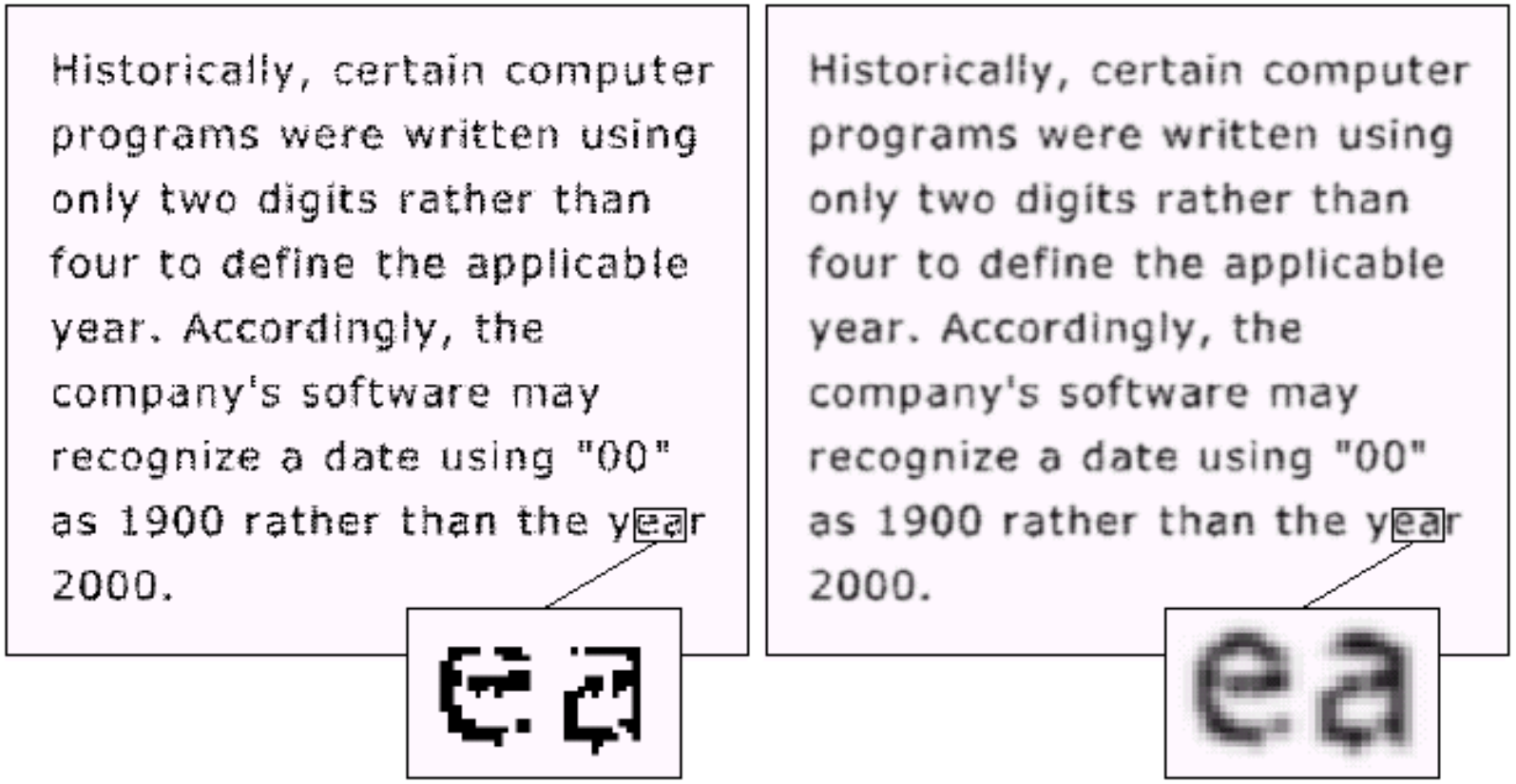
© 2002 R. C. Gonzalez & R. E. Woods



Additional Examples of Lowpass Filtering

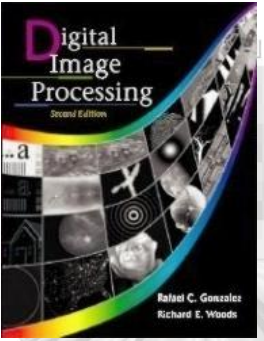
a b

FIGURE 4.19
(a) Sample text of poor resolution (note broken characters in magnified view). (b) Result of filtering with a GLPF (broken character segments were joined).



low pass filter احدى تطبيقاته

© 2002 R. C. Gonzalez & R. E. Woods



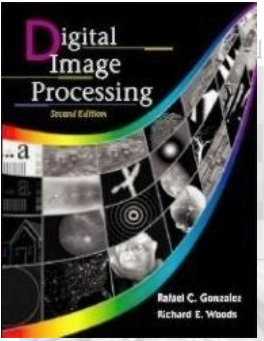
Additional Examples of Lowpass Filtering



a b c

FIGURE 4.20 (a) Original image (1028 × 732 pixels). (b) Result of filtering with a GLPF with $D_0 = 100$. (c) Result of filtering with a GLPF with $D_0 = 80$. Note reduction in skin fine lines in the magnified sections of (b) and (c).

© 2002 R. C. Gonzalez & R. E. Woods



Sharpening Frequency Domain Filter

$$H_{hp}(u,v) = H_{lp}(u,v)$$

Ideal highpass filter

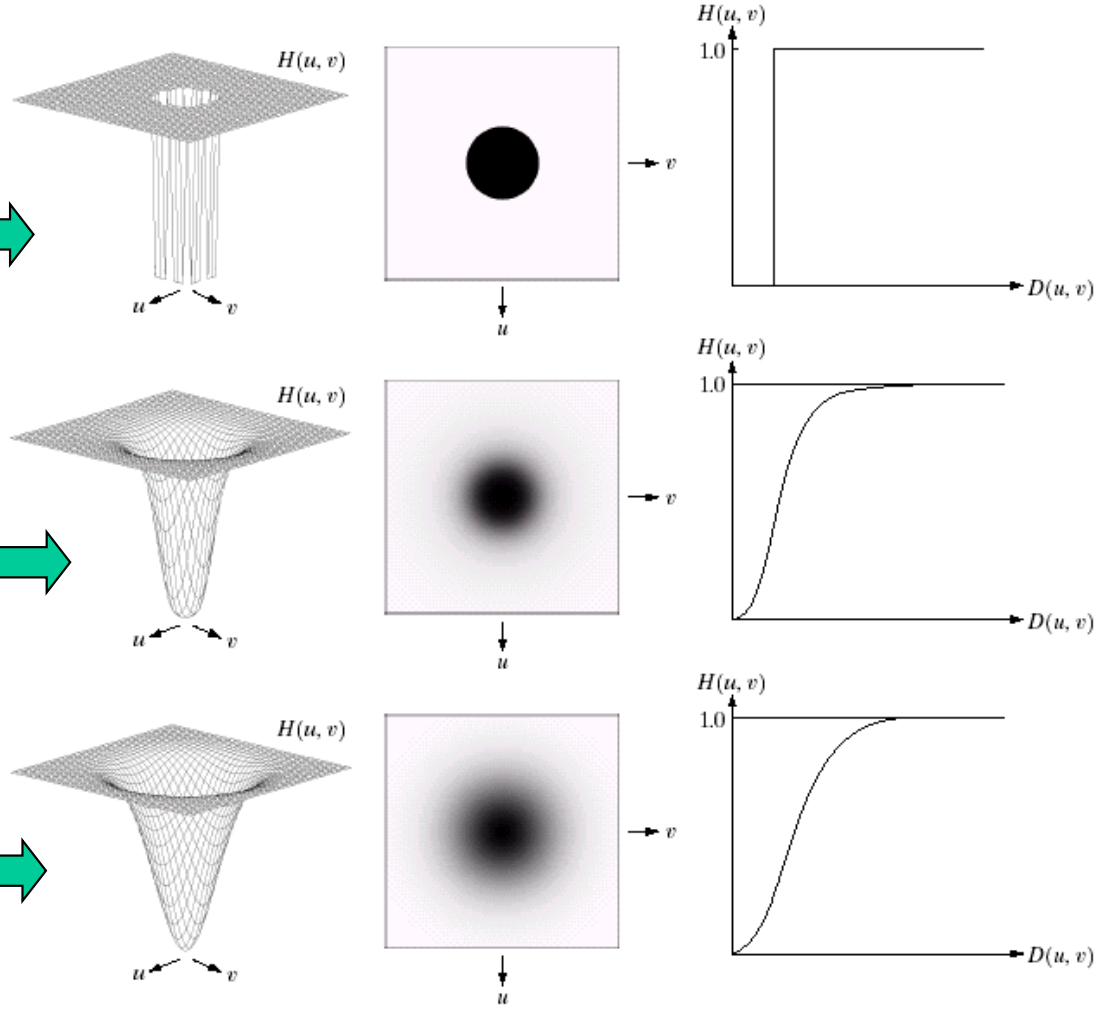
$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \le D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

Butterworth highpass filter

$$H(u,v) = \frac{1}{1+\left[D_0 / D(u,v)\right]^{2n}}$$

Gaussian highpass filter

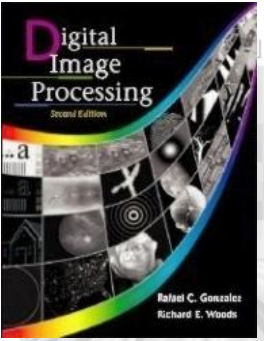
$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$



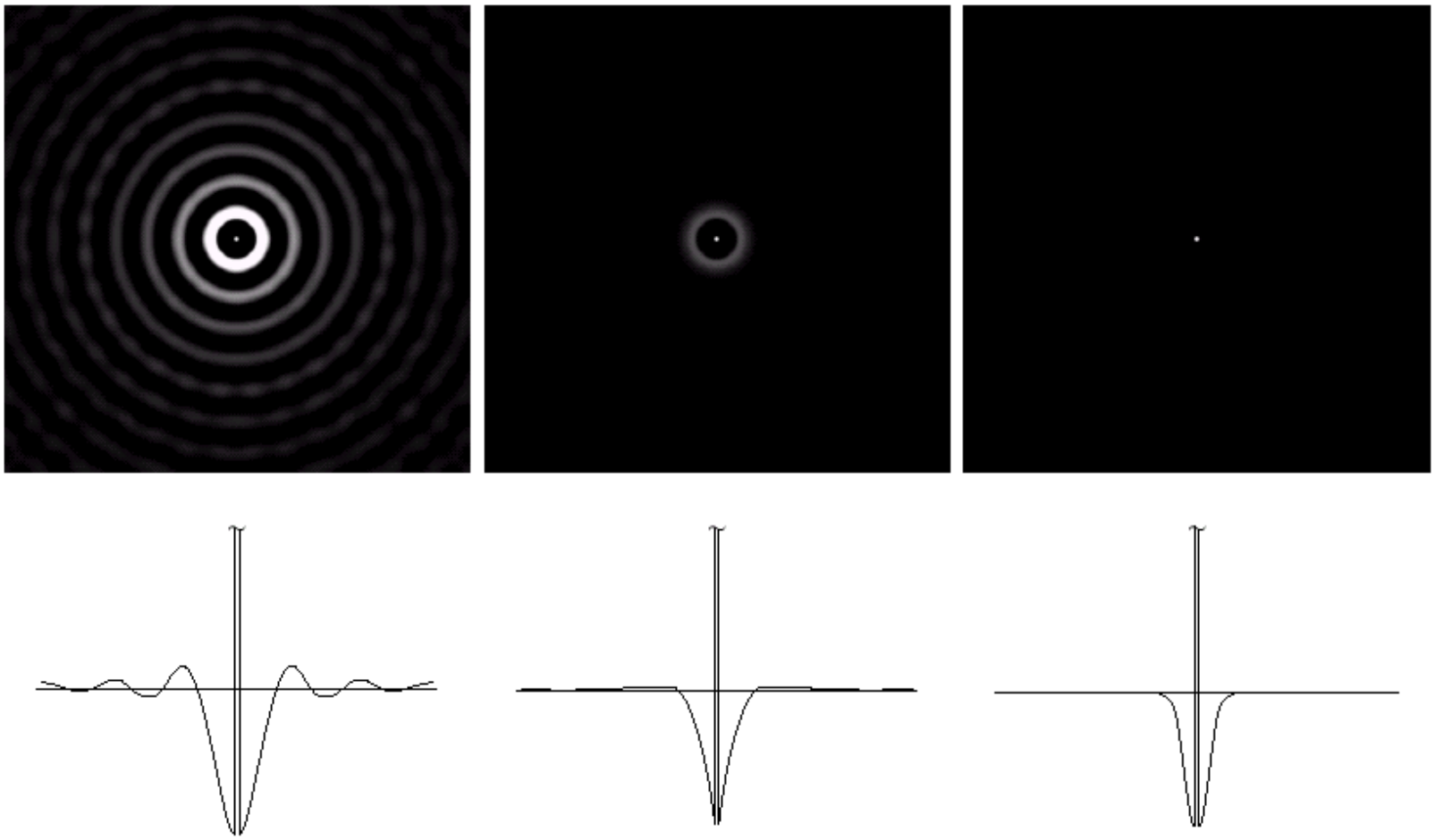
a b c
d e f
g h i

FIGURE 4.22 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

© 2002 R. C. Gonzalez & R. E. Woods



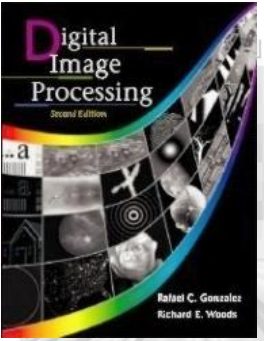
Highpass Filters
Spatial Representations



a b c

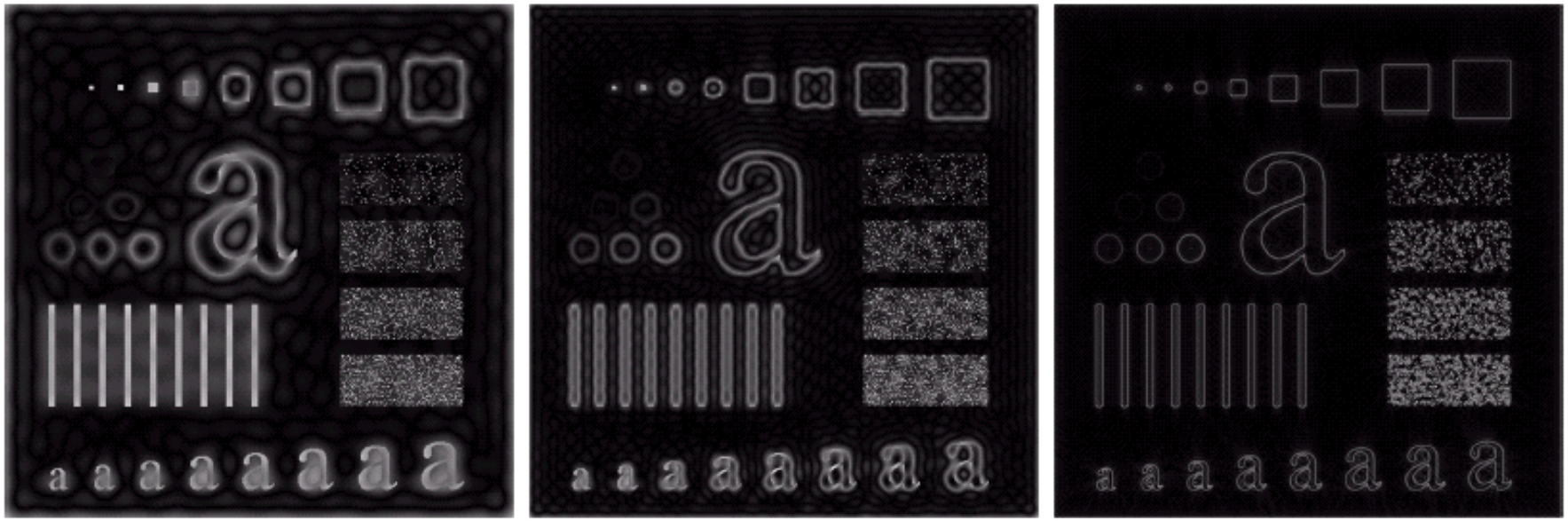
FIGURE 4.23 Spatial representations of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding gray-level profiles.

© 2002 R. C. Gonzalez & R. E. Woods



Ideal Highpass Filters

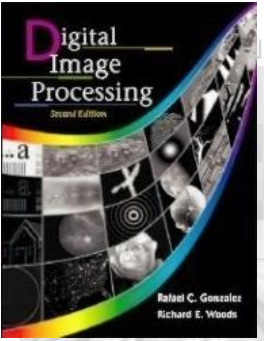
$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \le D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$



a b c

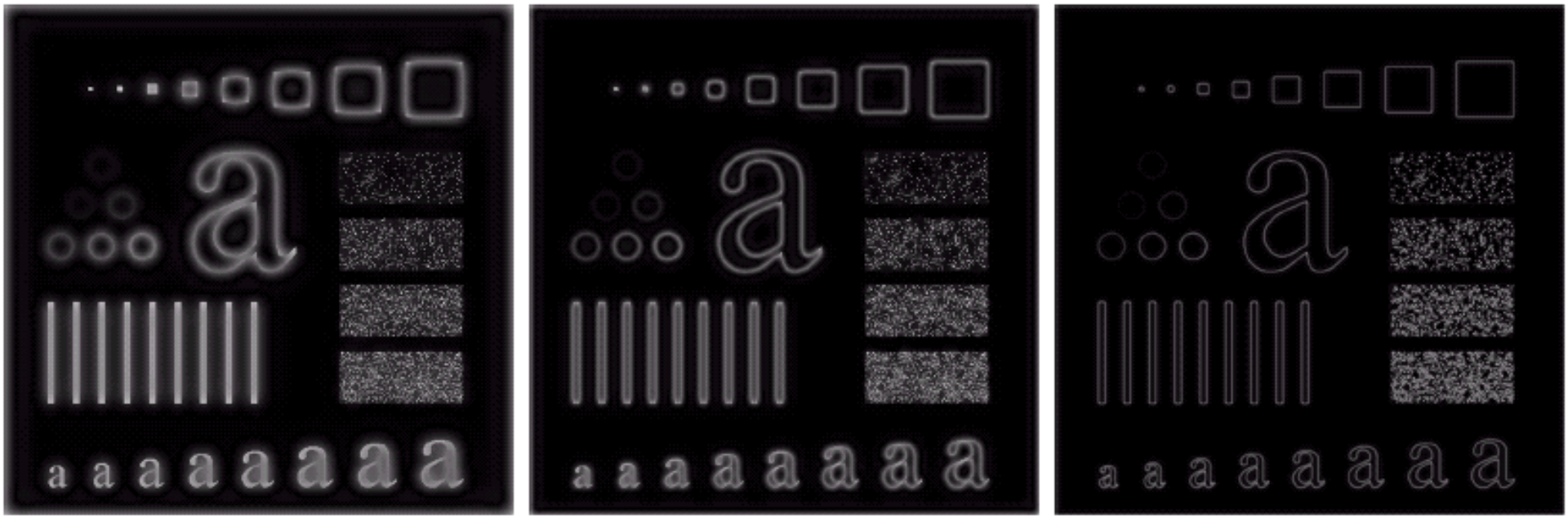
FIGURE 4.24 Results of ideal highpass filtering the image in Fig. 4.11(a) with D_0 = 15, 30, and 80, respectively. Problems with ringing are quite evident in (a) and (b).

© 2002 R. C. Gonzalez & R. E. Woods



Butterworth Highpass Filters

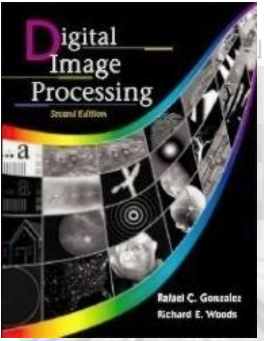
$$H(u,v) = \frac{1}{1+\left[D_0 / D(u,v)\right]^{2n}}$$



a b c

FIGURE 4.25 Results of highpass filtering the image in Fig. 4.11(a) using a BHPF of order 2 with D_0 = 15, 30, and 80, respectively. These results are much smoother than those obtained with an ILPF.

© 2002 R. C. Gonzalez & R. E. Woods



Gaussian Highpass Filters

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

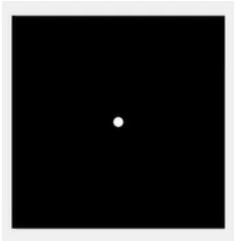


a b c

FIGURE 4.26 Results of highpass filtering the image of Fig. 4.11(a) using a GHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. Compare with Figs. 4.24 and 4.25.

© 2002 R. C. Gonzalez & R. E. Woods

```
%%%%% Ideal Low Pass Filter
i=imread('cameraman.tif');
fi=fftshift(fft2(i));
[m,n]=size(i);
[x,y]=meshgrid(1:n,1:m);
x=x-(n+1)/2;
y=y-(m+1)/2;
D=sqrt(x.^2+y.^2);
%%%%% Do=6 %%%%%%%%%
ILPF=D<6;
imshow(ILPF);
```



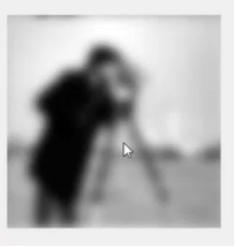
كلما كبرت هاي القيمة زادت عدد الـ Pixels اللي بتنحط تمر

```
%%%%% Ideal Low Pass Filter
i=imread('cameraman.tif');
fi=fftshift(fft2(i));
[m,n]=size(i);
[x,y]=meshgrid(1:n,1:m);
x=x-(n+1)/2;
y=y-(m+1)/2;
D=sqrt(x.^2+y.^2);
%%%%% Do=6 %%%%%%%%%
ILPF=D<6;
imshow(ILPF);
i_filt=fi.*ILPF;
ii= ifft2(fftshift(i_filt));
imshow(abs(ii),[]);
```



```
%%%%%% Gaussian Low Pass Filter (Do=6) %%%%%
Do=6;
GLPF=exp(-D.^2/(2*Do^2));
imshow(GLPF);
i_filtG=fi.*GLPF;
iiG= ifft2(fftshift(i_filtG));
imshow(abs(iiG),[]);
```

Do=6



وكلما زادت قيمة Do زاد الـ smoothness اللي

Do=20



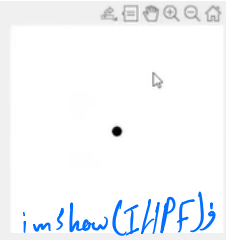
```
%%%%%% Butterworth Low Pass Filter (Do=6) %%%%%
Do=6;
n=2;
BLPF=1./(1.0+(D./Do).^(2*n));
imshow(BLPF);
i_filtB=fi.*BLPF;
iiB= ifft2(fftshift(i_filtB));
imshow(abs(iiB),[]);
```



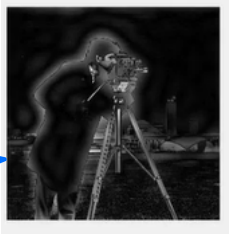


كلما زادت قيمة n زاد عدد الـ pixels المسموح بمرورها

```
%%%%%%%%% High Pass Filter %%%%%%
i=imread('cameraman.tif');
fi=fftshift(fft2(i));
[m,n]=size(i);
[x,y]=meshgrid(1:n,1:m);
x=x-(n+1)/2;
y=y-(m+1)/2;
D=sqrt(x.^2+y.^2);
%%%%% Ideal High Pass Filter (Do=6) %%%%%%
IHPF=D>6;
imshow(IHPF);
i_filt=fi.*IHPF;
ii= ifft2(fftshift(i_filt));
imshow(abs(ii),[]);
%%%%%% Gaussian High Pass Filter (Do=6) %%%%%
Do=6;
GHPF=1-(exp(-D.^2/(2*Do^2)));
imshow(GHPF);
i_filtG=fi.*GHPF;
iiG= ifft2(fftshift(i_filtG));
imshow(abs(iiG),[]);
%%%%%% Butterworth High Pass Filter (Do=6) %%%%%
Do=6;
n=1;
BHPF=1./(1.0+(Do./D).^(2*n));
imshow(BHPF);
i_filtB=fi.*BHPF;
iiB= ifft2(fftshift(i_filtB));
imshow(abs(iiB),[]);
```

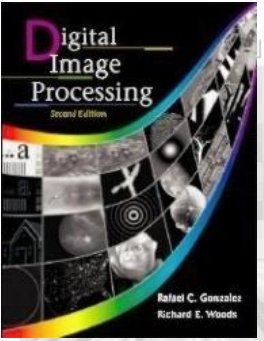


imshow(IHPF)





ما تزيد قيمة D تقل عدد ال Pixels المسموح المرور لها.



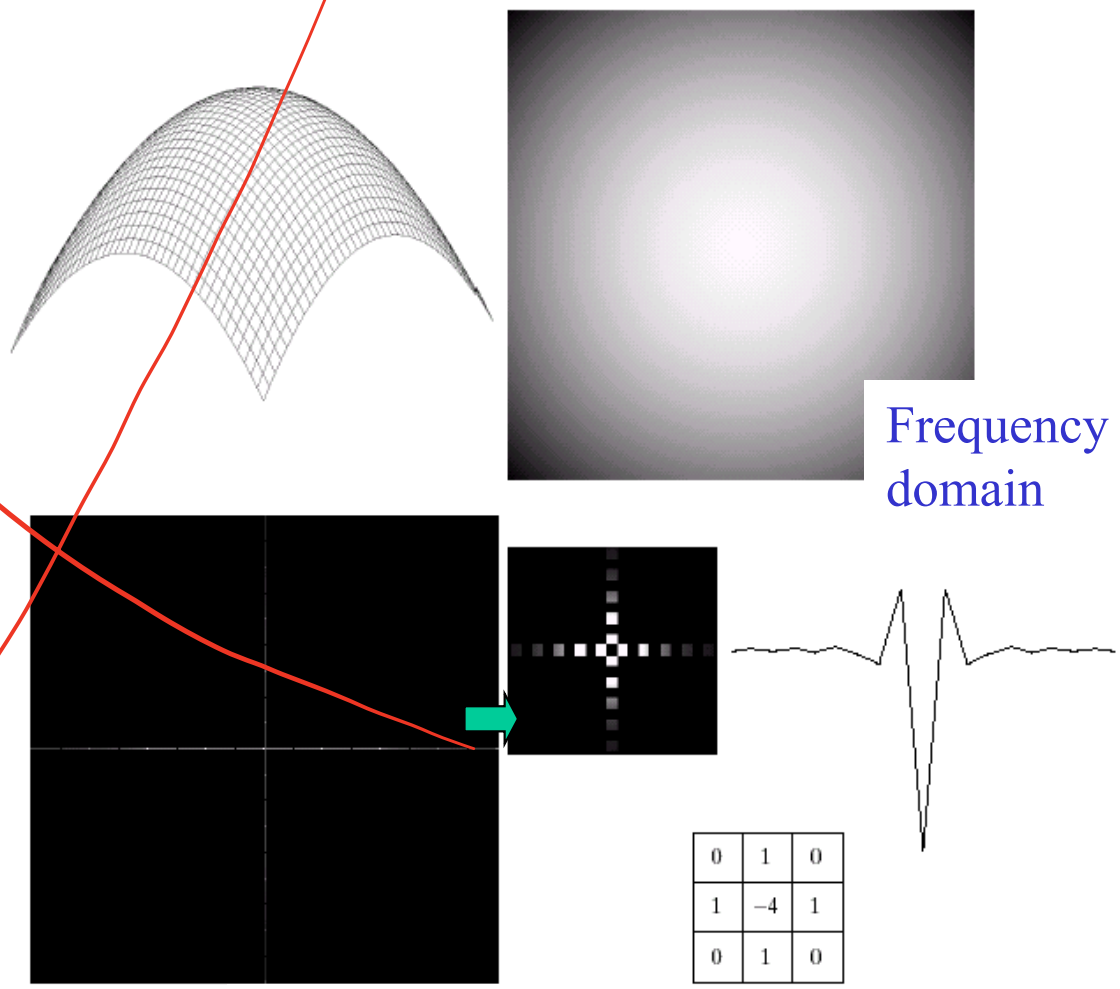
The Laplacian in the Frequency Domain

- The Laplacian filter

$$H(u,v) = -(u^2 + v^2)$$

- Shift the center:

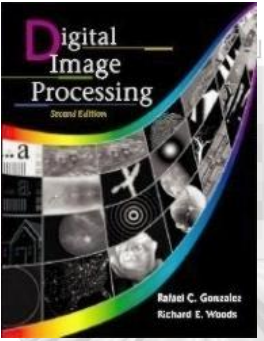
$$H(u,v) = -\left[(u - \frac{M}{2})^2 + (v - \frac{N}{2})^2\right]$$



a b
c d e
f

FIGURE 4.27 (a) 3-D plot of Laplacian in the frequency domain. (b) Image representation of (a). (c) Laplacian in the spatial domain obtained from the inverse DFT of (b). (d) Zoomed section of the origin of (c). (e) Gray-level profile through the center of (d). (f) Laplacian mask used in Section 3.7.

© 2002 R. C. Gonzalez & R. E. Woods



a b
c d

FIGURE 4.28
(a) Image of the North Pole of the moon.
(b) Laplacian filtered image.
(c) Laplacian image scaled.
(d) Image enhanced by using Eq. (4.4-12).
(Original image courtesy NA

$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

where

$\nabla^2 f(x,y)$: the Laplacian-filtered image in the spatial domain

For display purposes only

© 2002 R. C. Gonzalez & R. E. Woods